

BioETL Foundation Diagrams with Descriptions

Total diagrams: 54

Source folder: docs/02-architecture/mmd-diagrams/foundation

Description folder: docs/02-architecture/diagram-descriptions/mmd-diagrams/foundation

Generated: 2026-03-02T13:00:13+03:00

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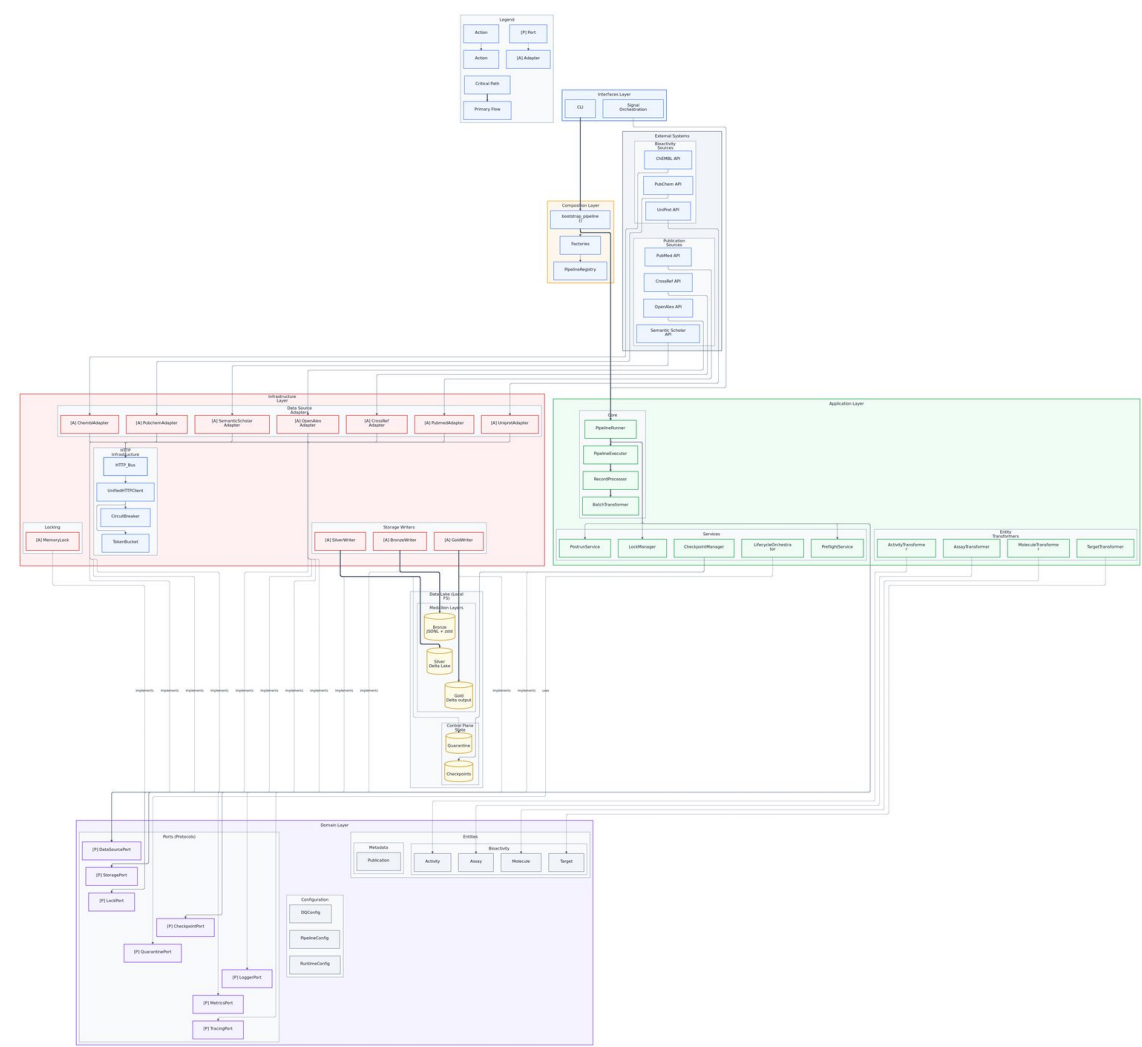
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01. Title: Full System Component Diagram

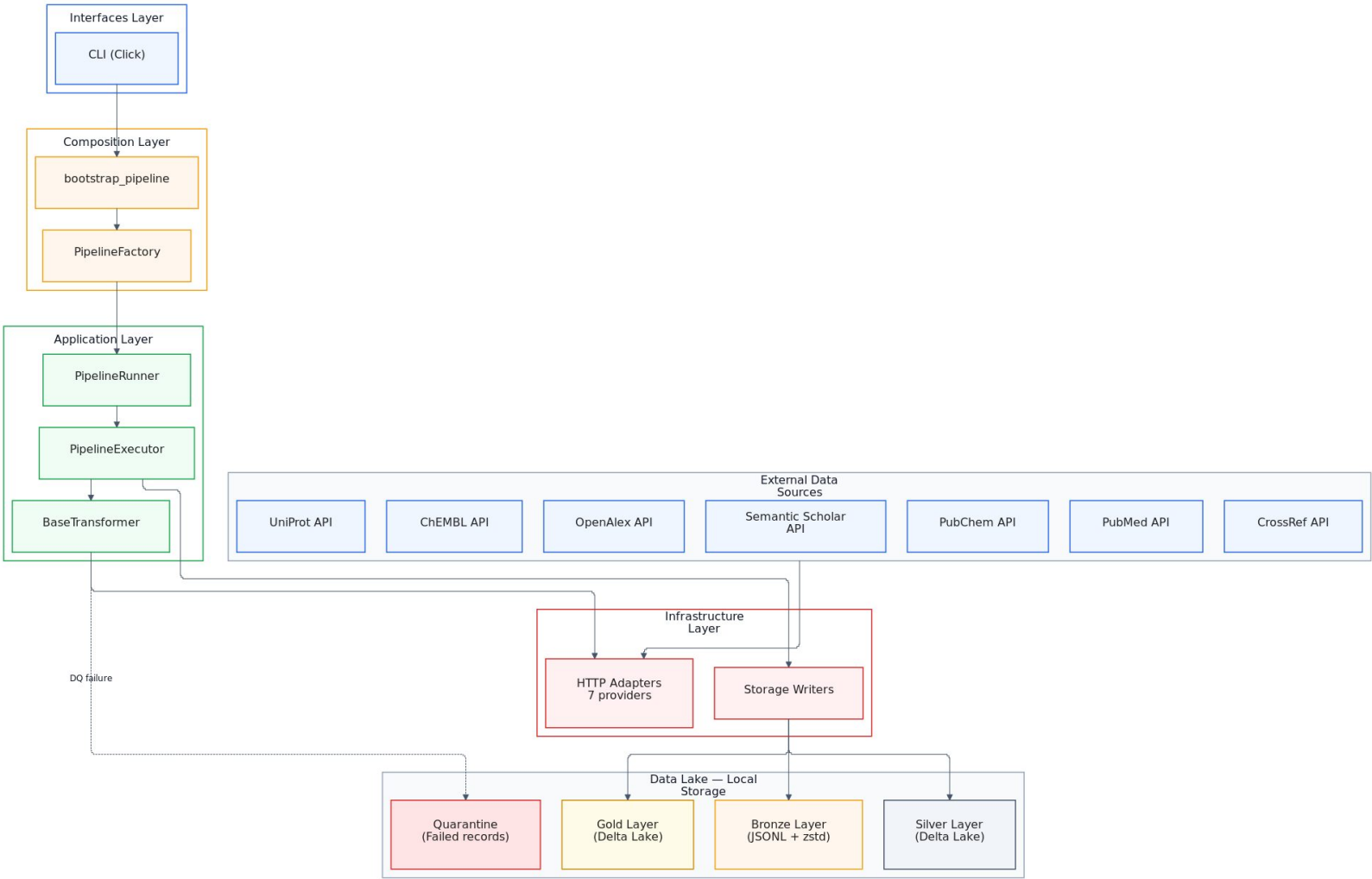
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02. Title: High-Level System Architecture

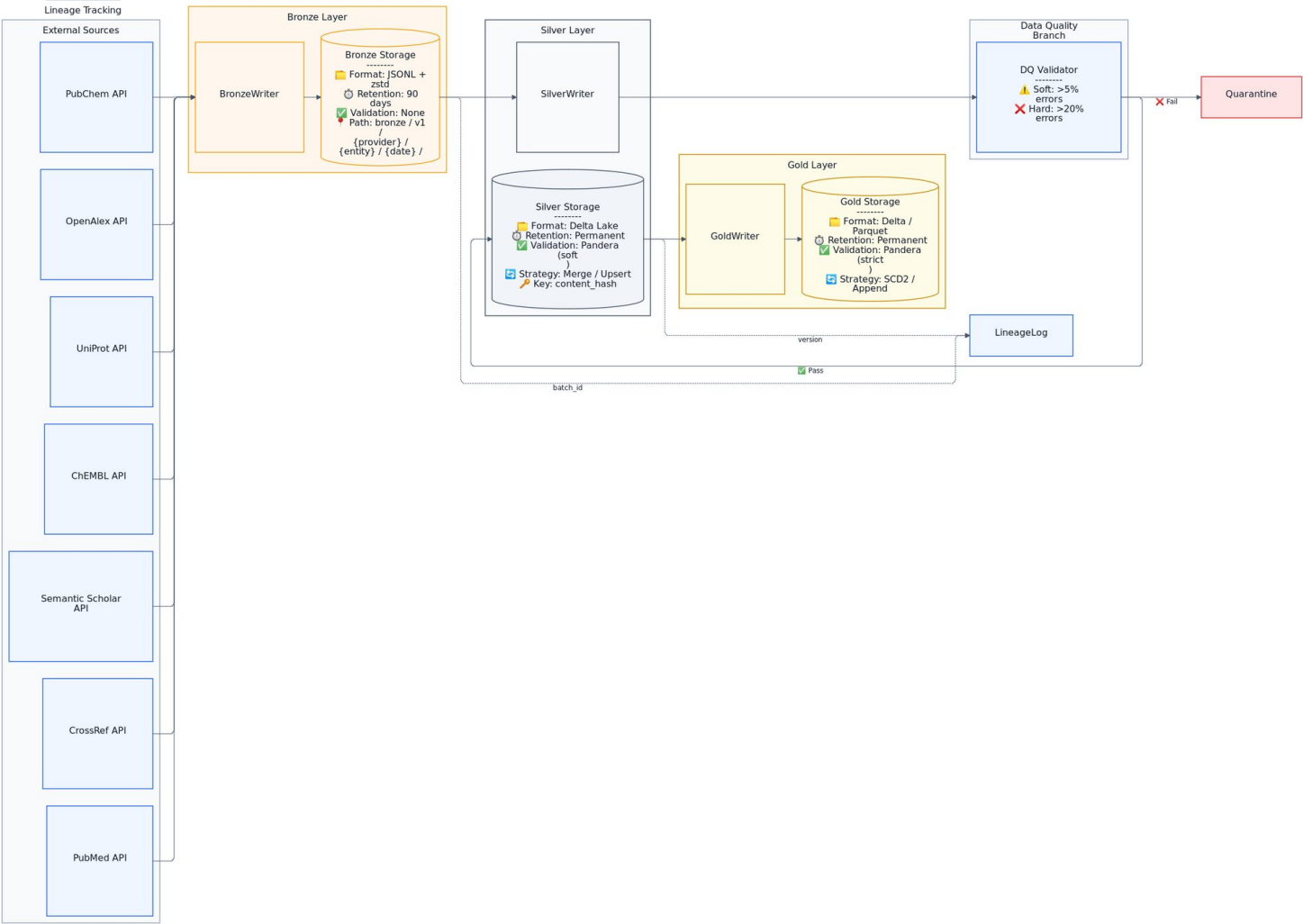
ID: 01-high-level Type: flowchart Level: Mixed (System / Component / Class) Date: 2026-02-24

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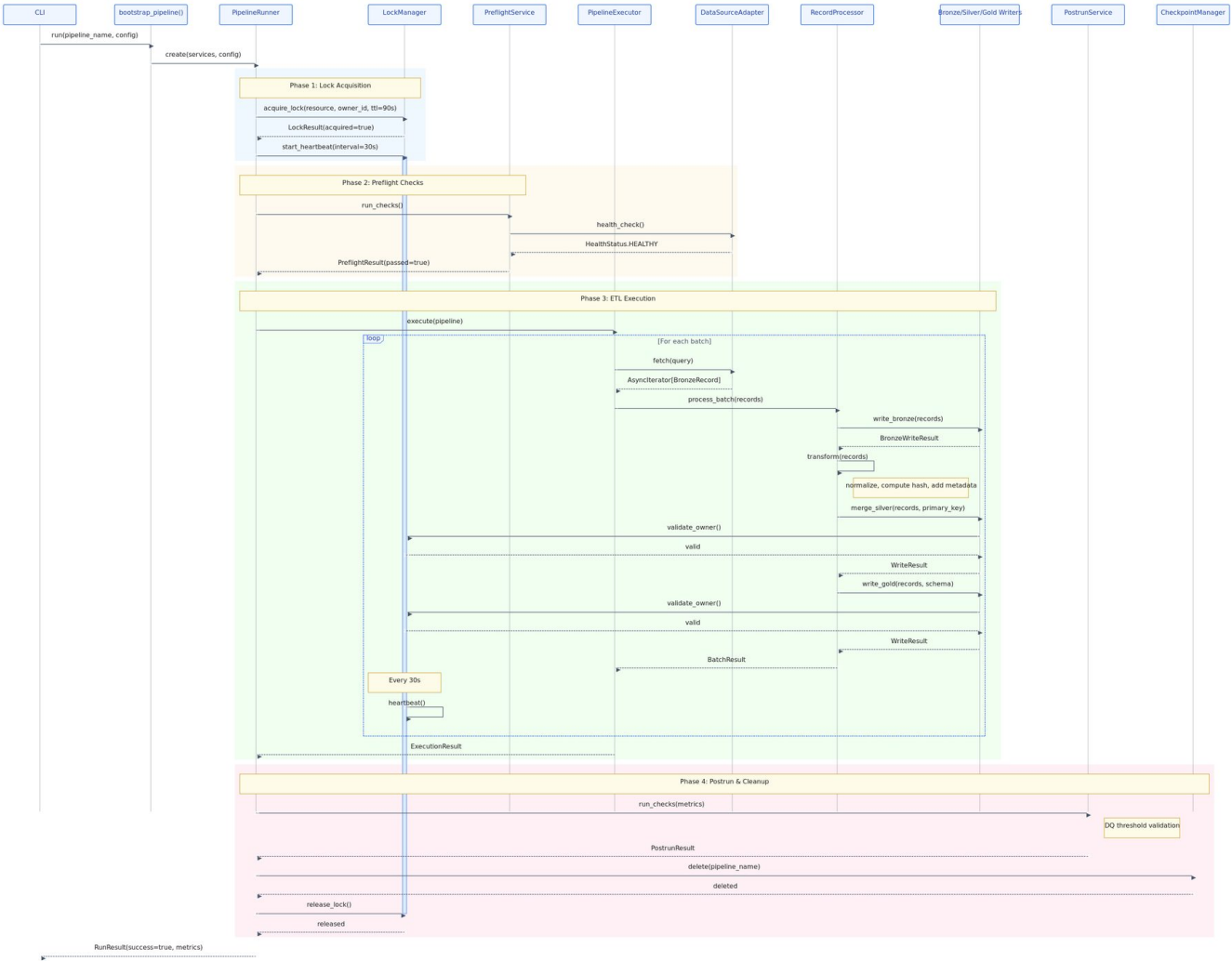
03. Title: Full Medallion Data Flow with Lineage and DQ

ID: 02-full-medallion-data-flow Type: flowchart Level: Mixed (System / Component / Class) Date: 2026-02-24



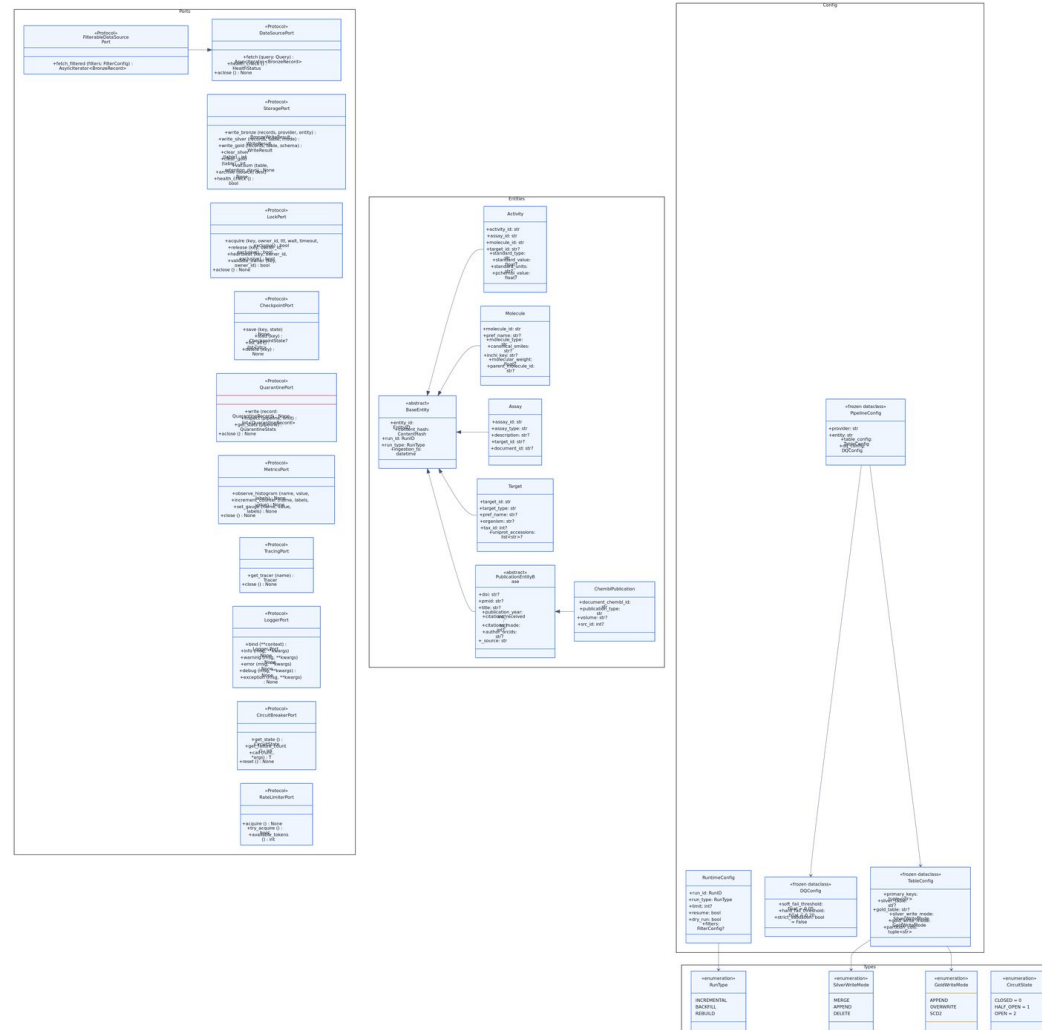
04. Title: Pipeline Execution — Happy Path

ID: 03-pipeline-execution-happy-path Type: sequenceDiagram Level: Mixed (System / Component / Class) Date: 2026-02-24



05. Title: Domain Layer Class Diagram

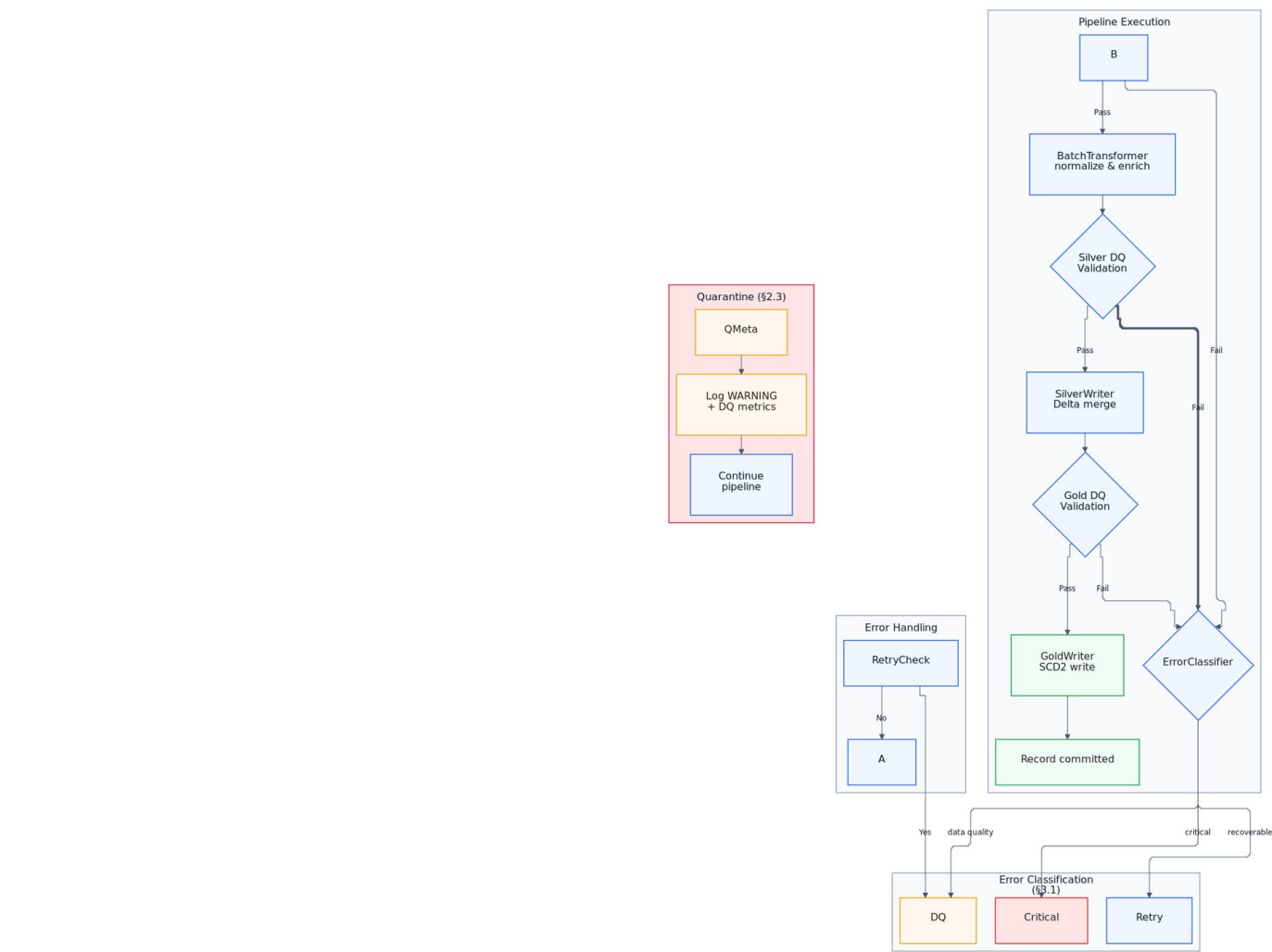
ID: 04-domain-layer-class-diagram Type: classDiagram Level: Mixed (System / Component / Class) Date: 2026-02-24



06. Title: Error Handling and Quarantine Flow

ID: 04-error-flow Type: flowchart Level: Mixed (System / Component / Class) Date: 2026-02-24

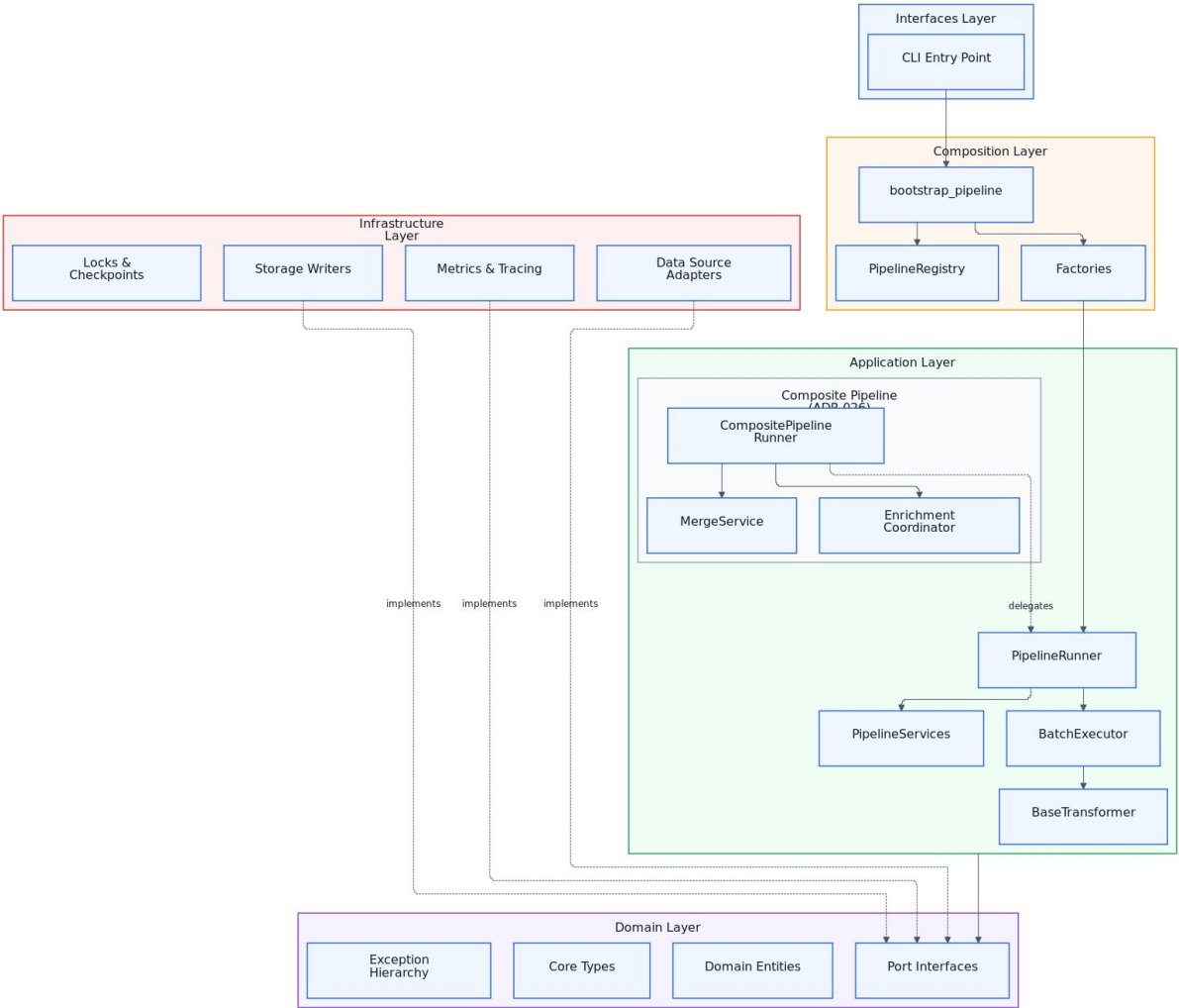
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07. Title: Layer Interaction — Hexagonal Architecture

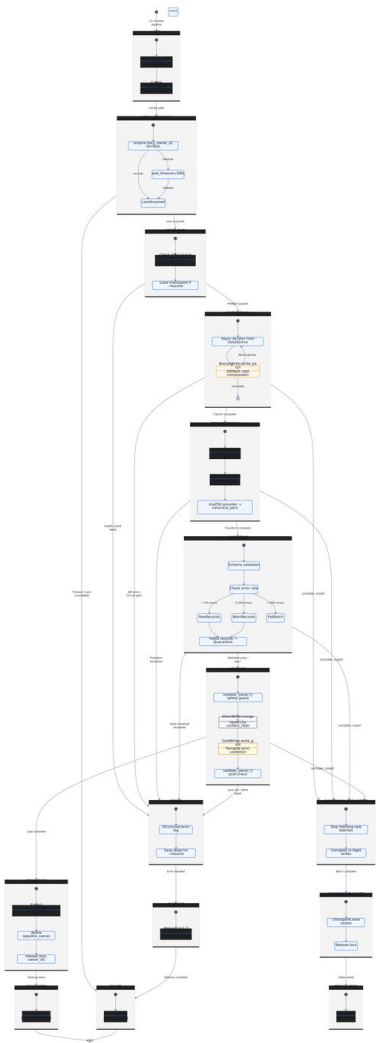
ID: 05-layers-interaction Type: flowchart Level: Mixed (System / Component / Class) Date: 2026-02-24

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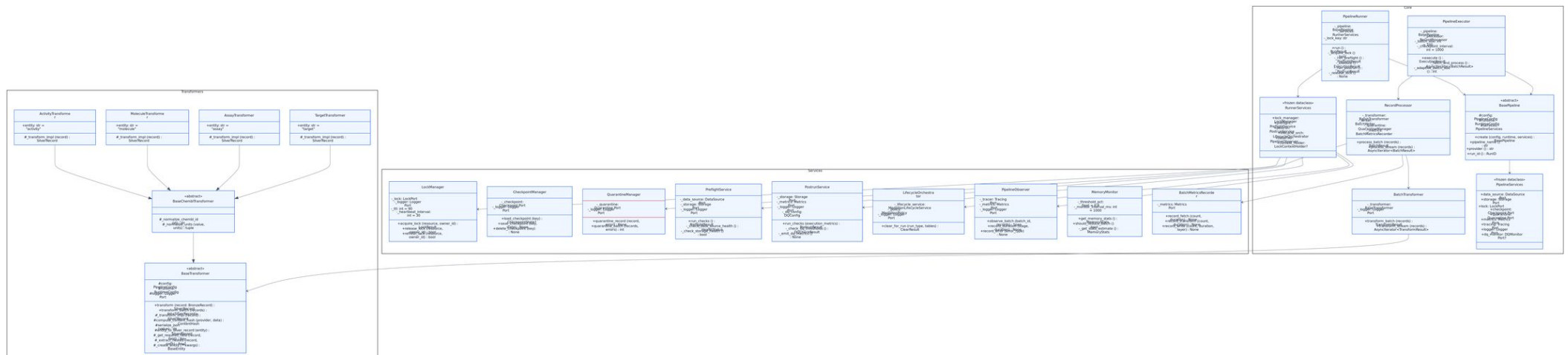
08. Title: Pipeline Lifecycle State Machine

ID: 05-pipeline-lifecycle-states Type: stateDiagram Level: Mixed (System / Component / Class) Date: 2026-02-24



09. Title: Application Layer Class Diagram

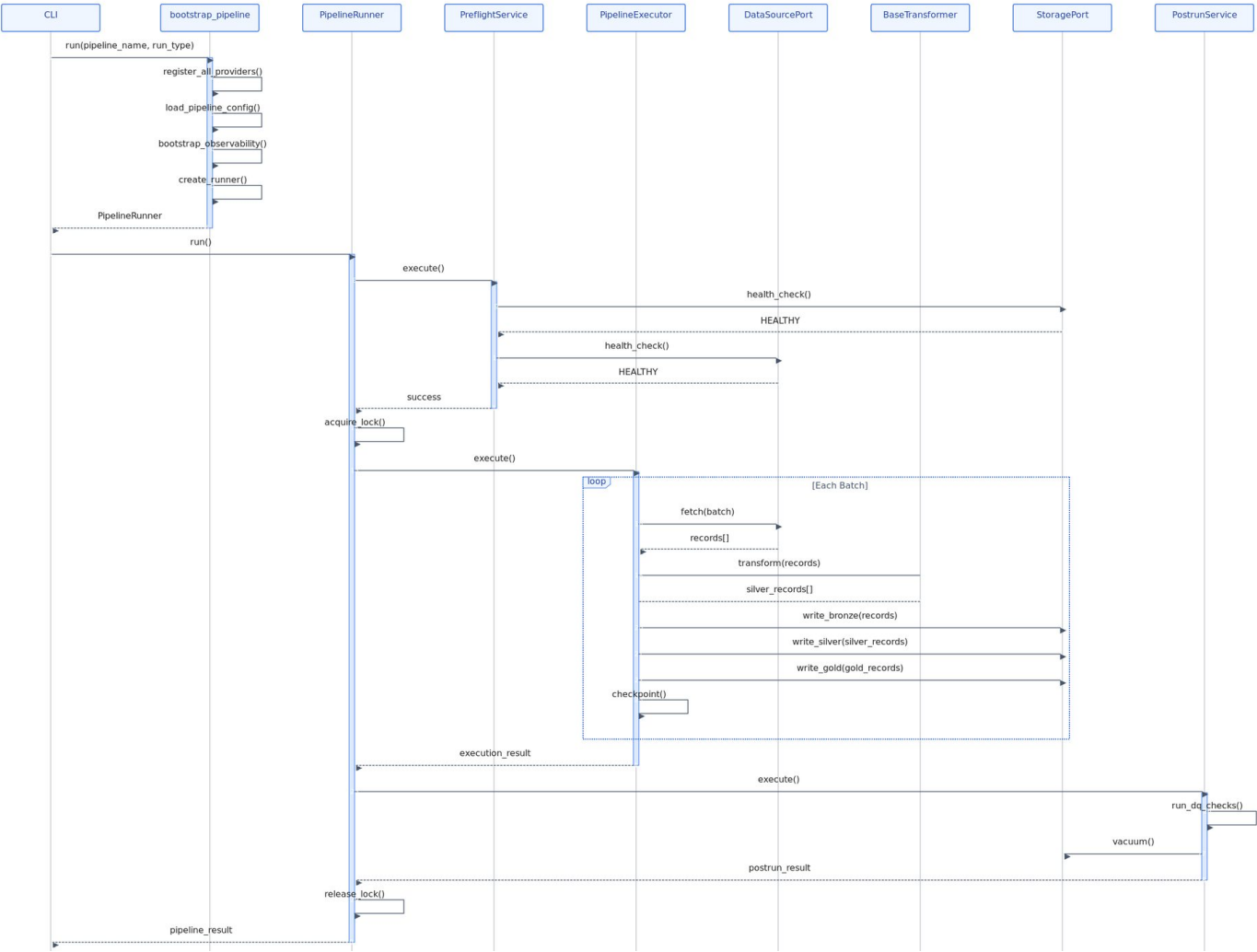
ID: 06-application-layer-class-diagram Type: classDiagram Level: Mixed (System / Component / Class) Date: 2026-02-24



10. Title: Pipeline Execution Sequence (Full)

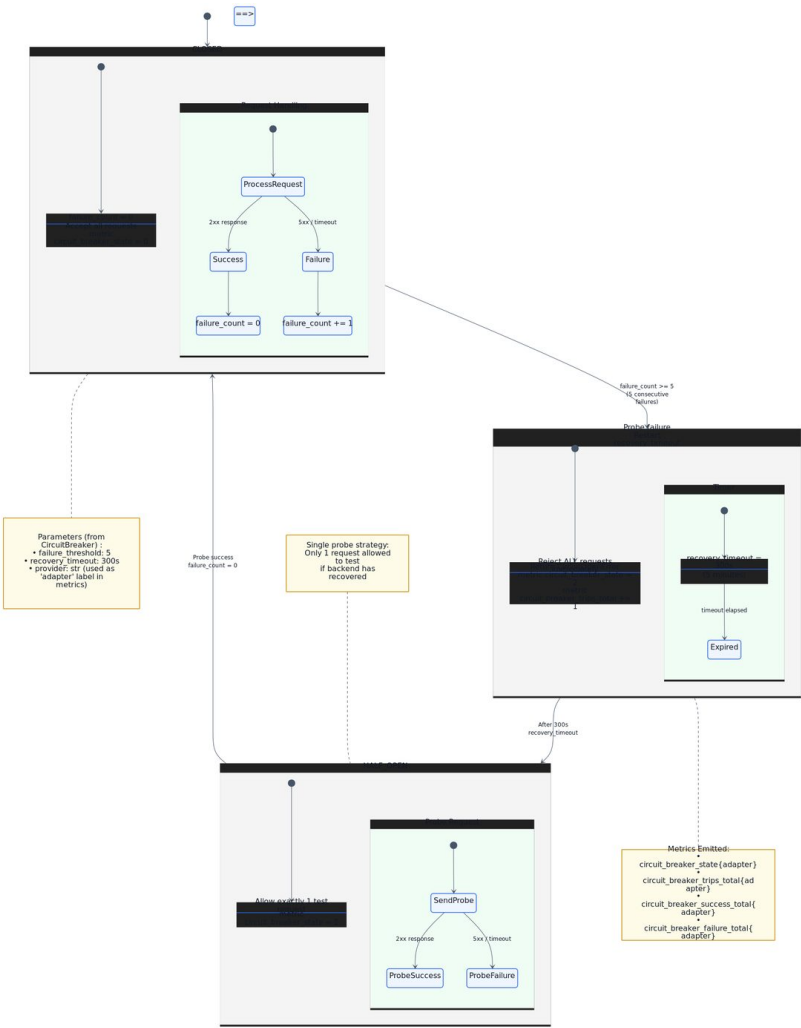
ID: 06-pipeline-execution Type: sequenceDiagram Level: Mixed (System / Component / Class) Date: 2026-02-24

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11. Title: Circuit Breaker State Machine

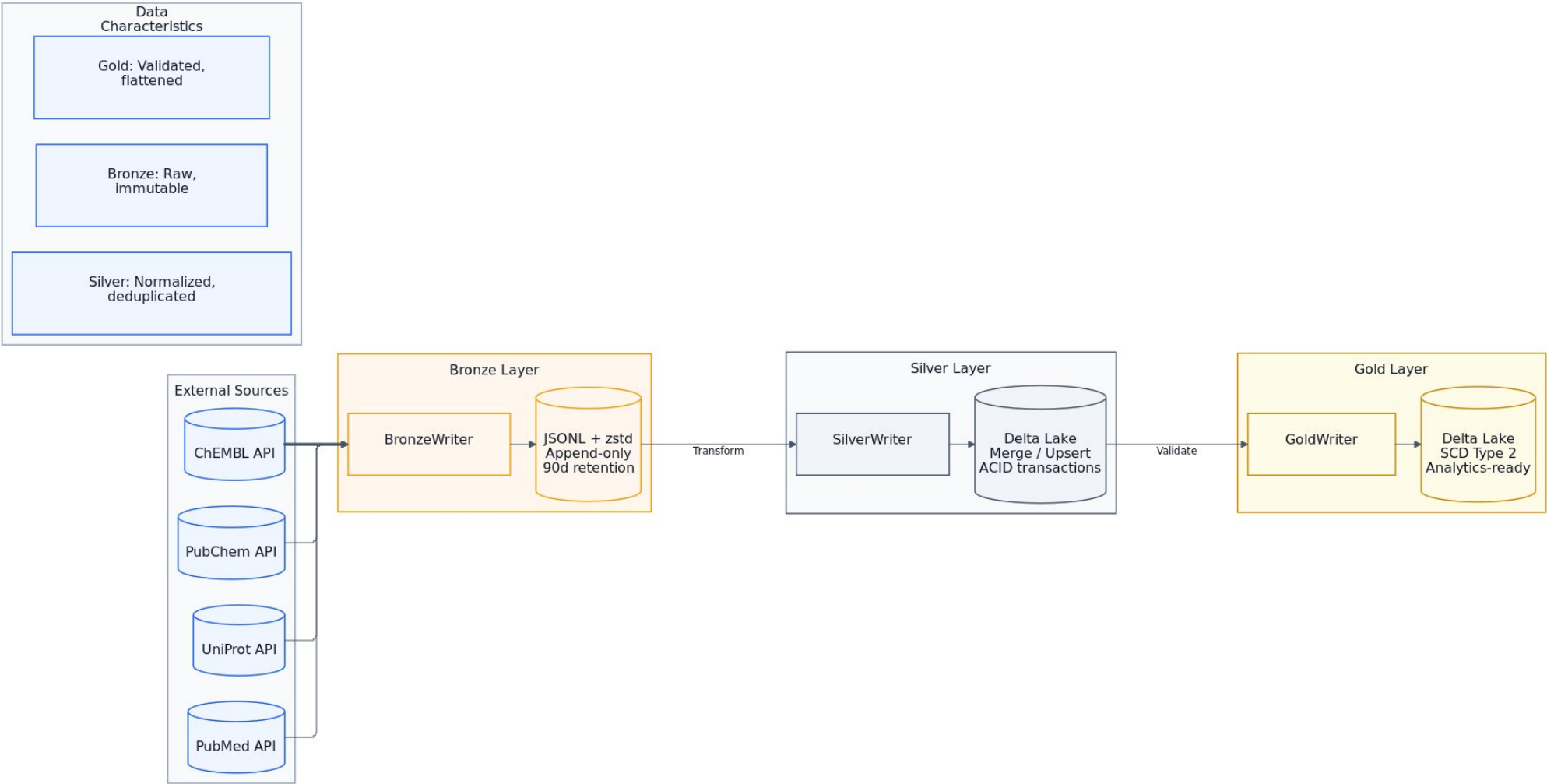
ID: 07-circuit-breaker-states Type: stateDiagram Level: Mixed (System / Component / Class) Date: 2026-02-24



12. Title: Medallion Data Flow (Sources → Bronze → Silver → Gold)

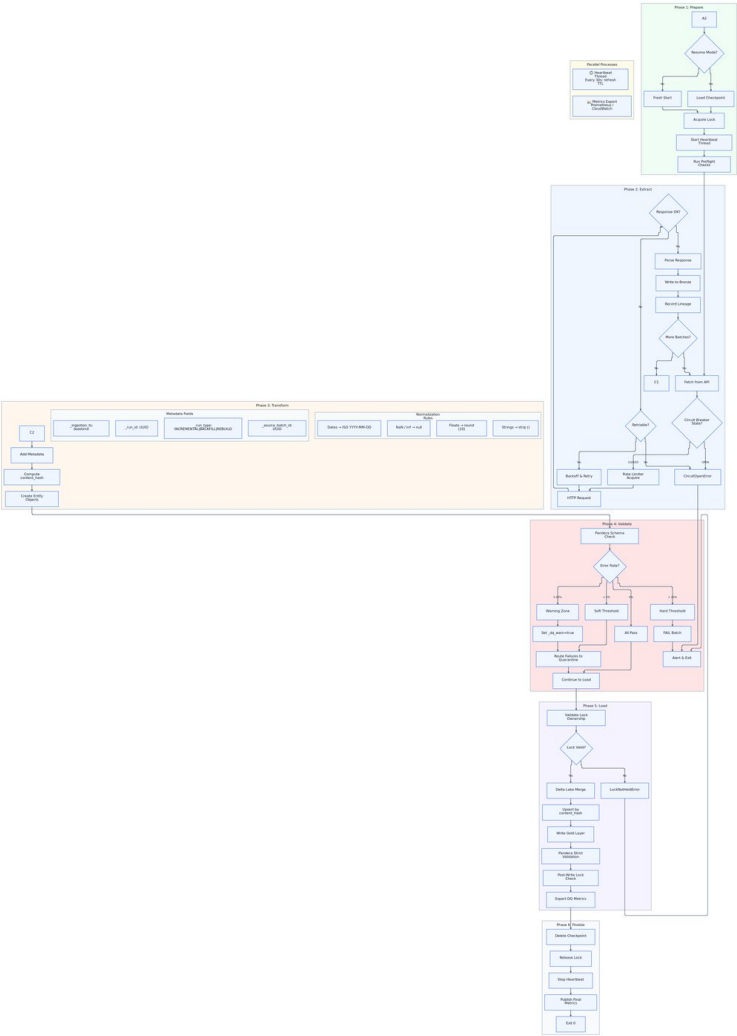
ID: 07-medallion-flow Type: flowchart Level: Mixed (System / Component / Class) Date: 2026-02-24

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13. Title: Complete ETL Workflow (6 Phases)

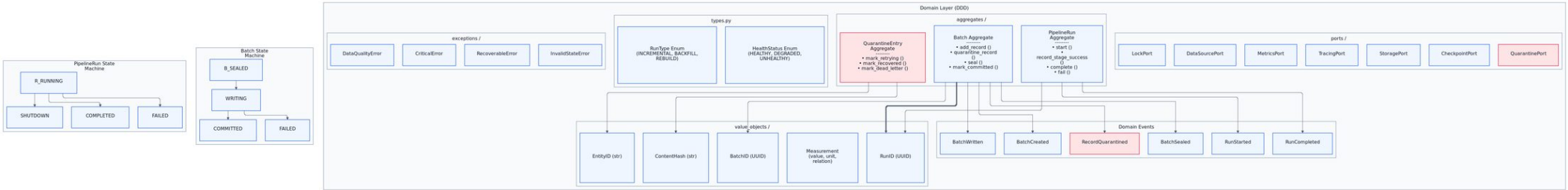
ID: 08-complete-etl-workflow Type: flowchart Level: Mixed (System / Component / Class) Date: 2026-02-24



14. Title: Domain Layer — DDD Components

ID: 08-domain-ddd Type: flowchart Level: Mixed (System / Component / Class) Date: 2026-02-24

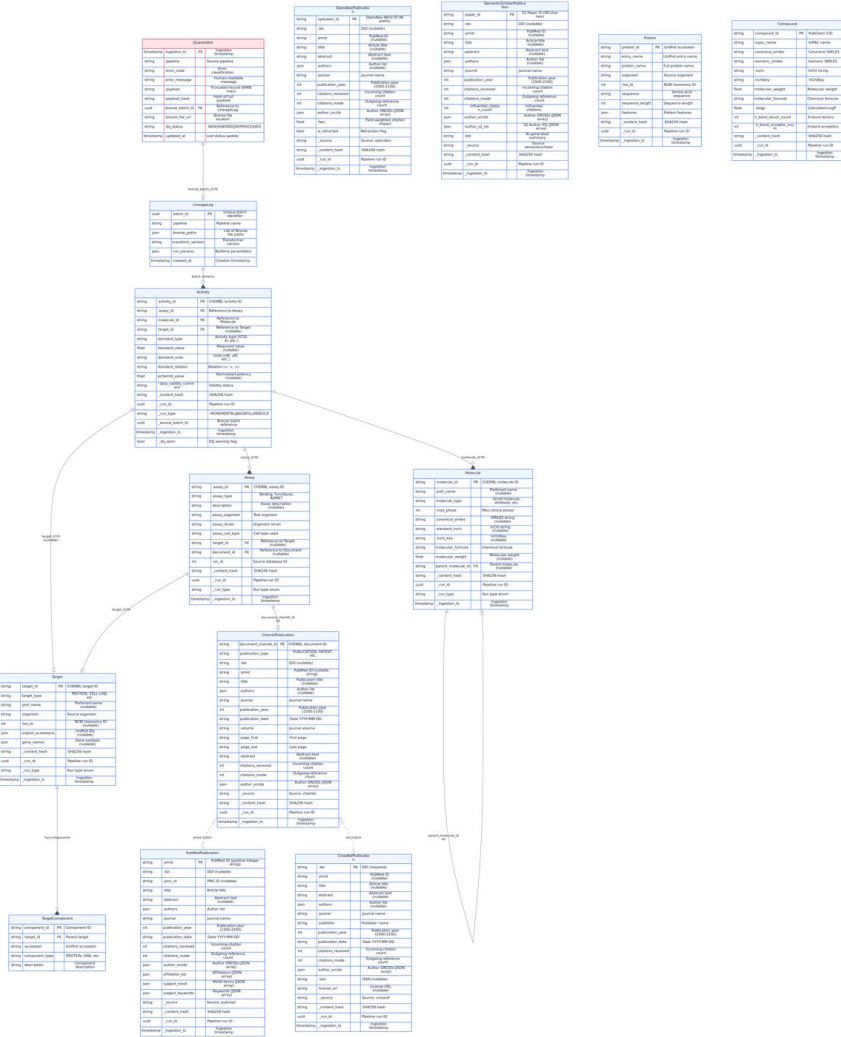
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15. Title: Entity-Relationship Diagram (All Providers)

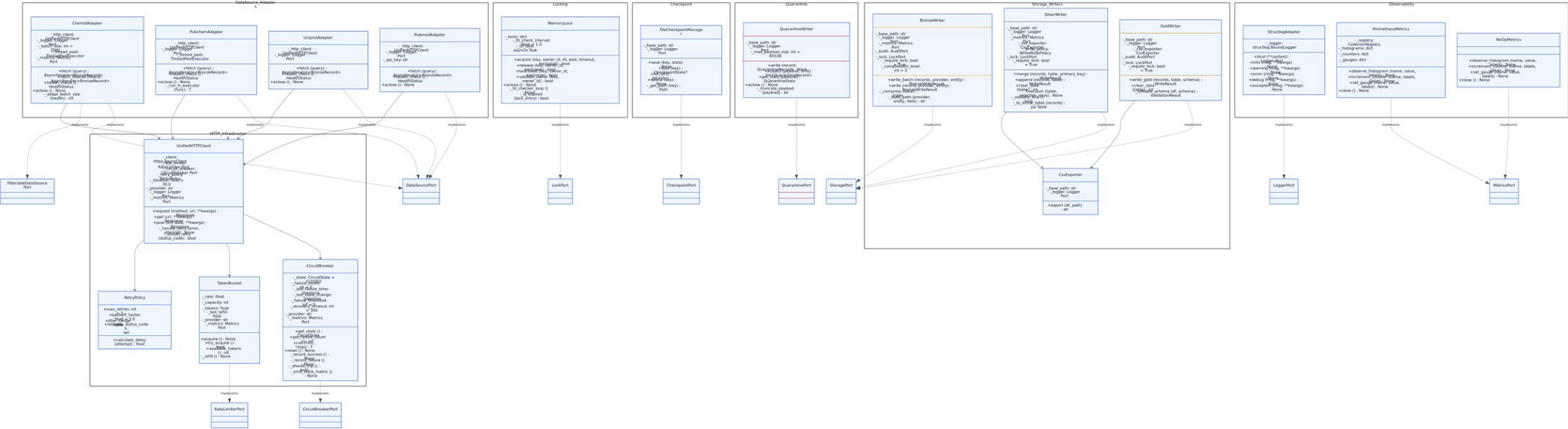
ID: 09-full-er-diagram Type: erDiagram Level: Mixed (System / Component / Class) Date: 2026-02-24

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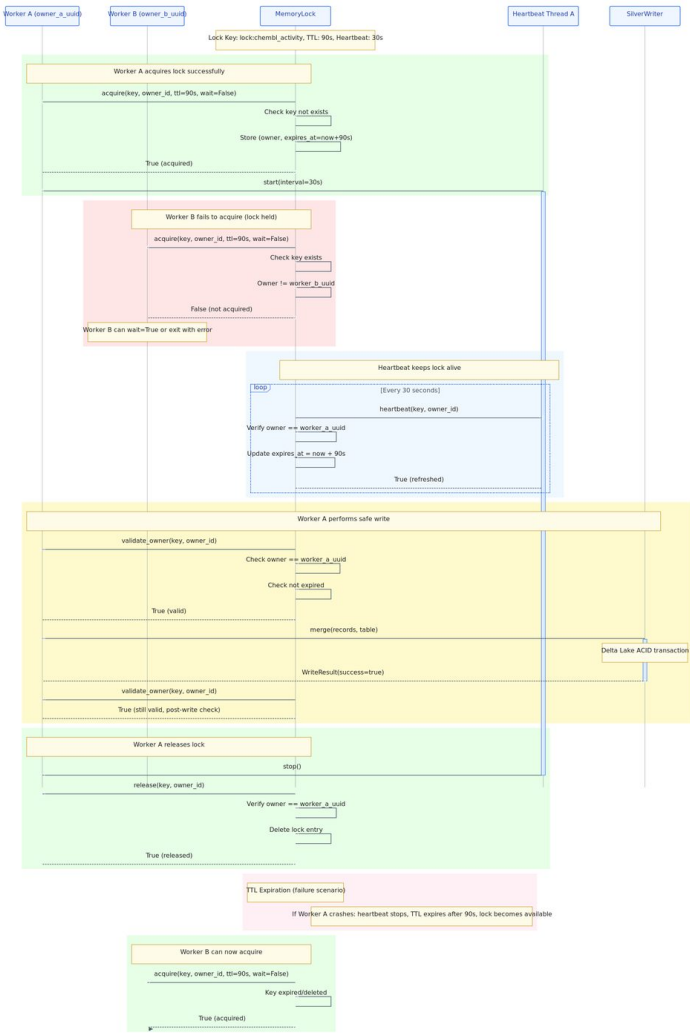
16. Title: Infrastructure Layer Class Diagram

ID: 10-infrastructure-layer-class-diagram Type: classDiagram Level: Mixed (System / Component / Class) Date: 2026-02-24



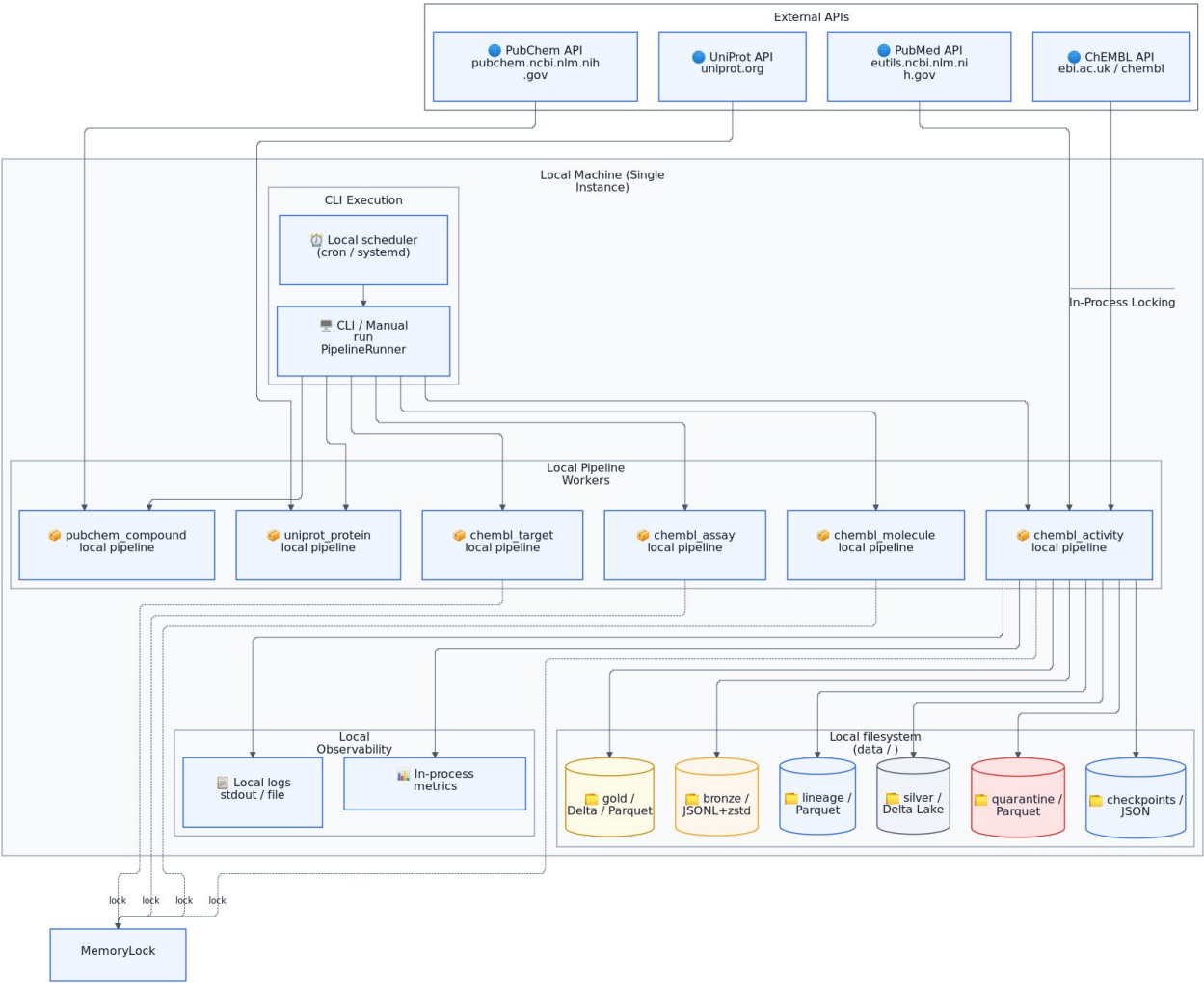
17. Title: Lock Acquisition Sequence (Two Workers)

ID: 11-lock-acquisition-sequence Type: sequenceDiagram Level: Mixed (System / Component / Class) Date: 2026-02-24



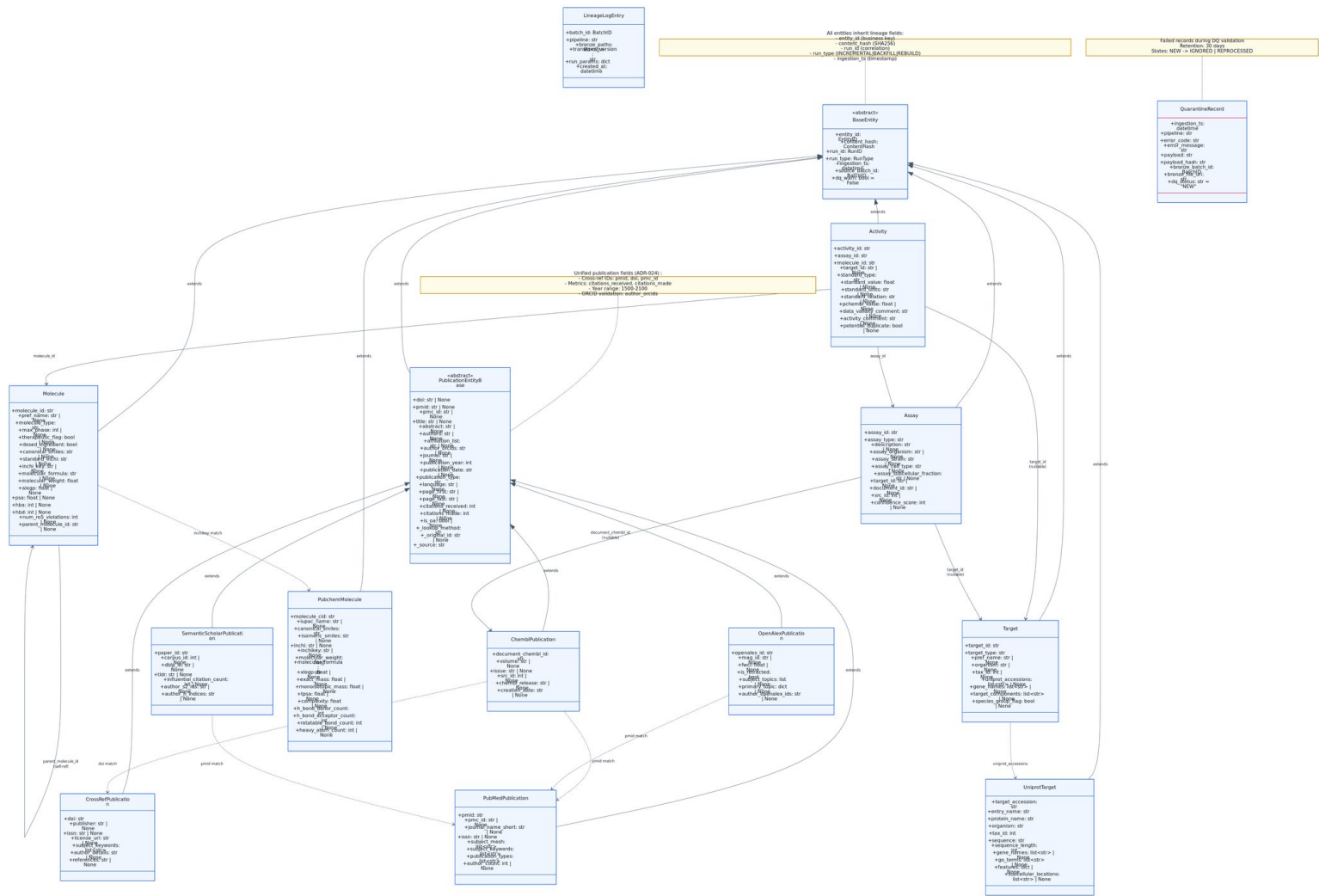
18. Title: Local Deployment Architecture

ID: 12-local-deployment-architecture Type: flowchart Level: Mixed (System / Component / Class) Date: 2026-02-24



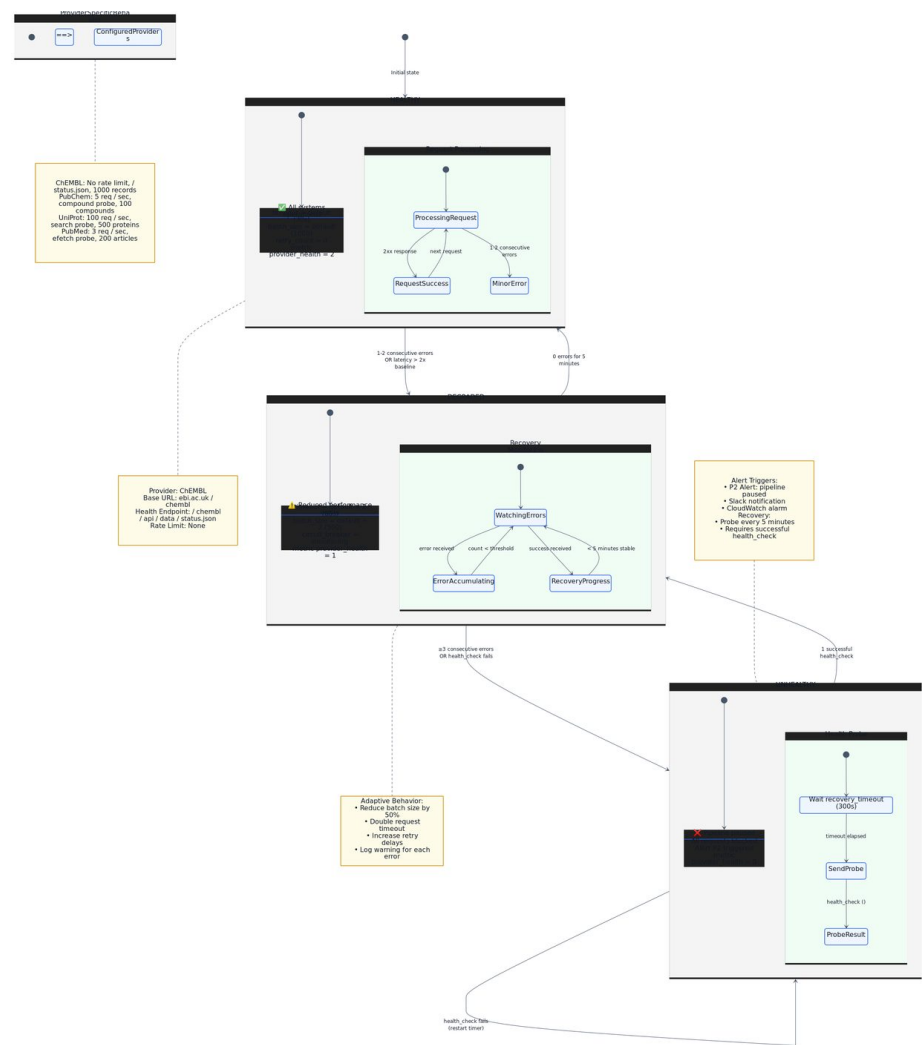
19. Title: Domain Models Relationship Hierarchy

ID: 13-domain-models-relationship Type: classDiagram Level: Mixed (System / Component / Class) Date: 2026-02-24



20. Title: Provider Health State Machine

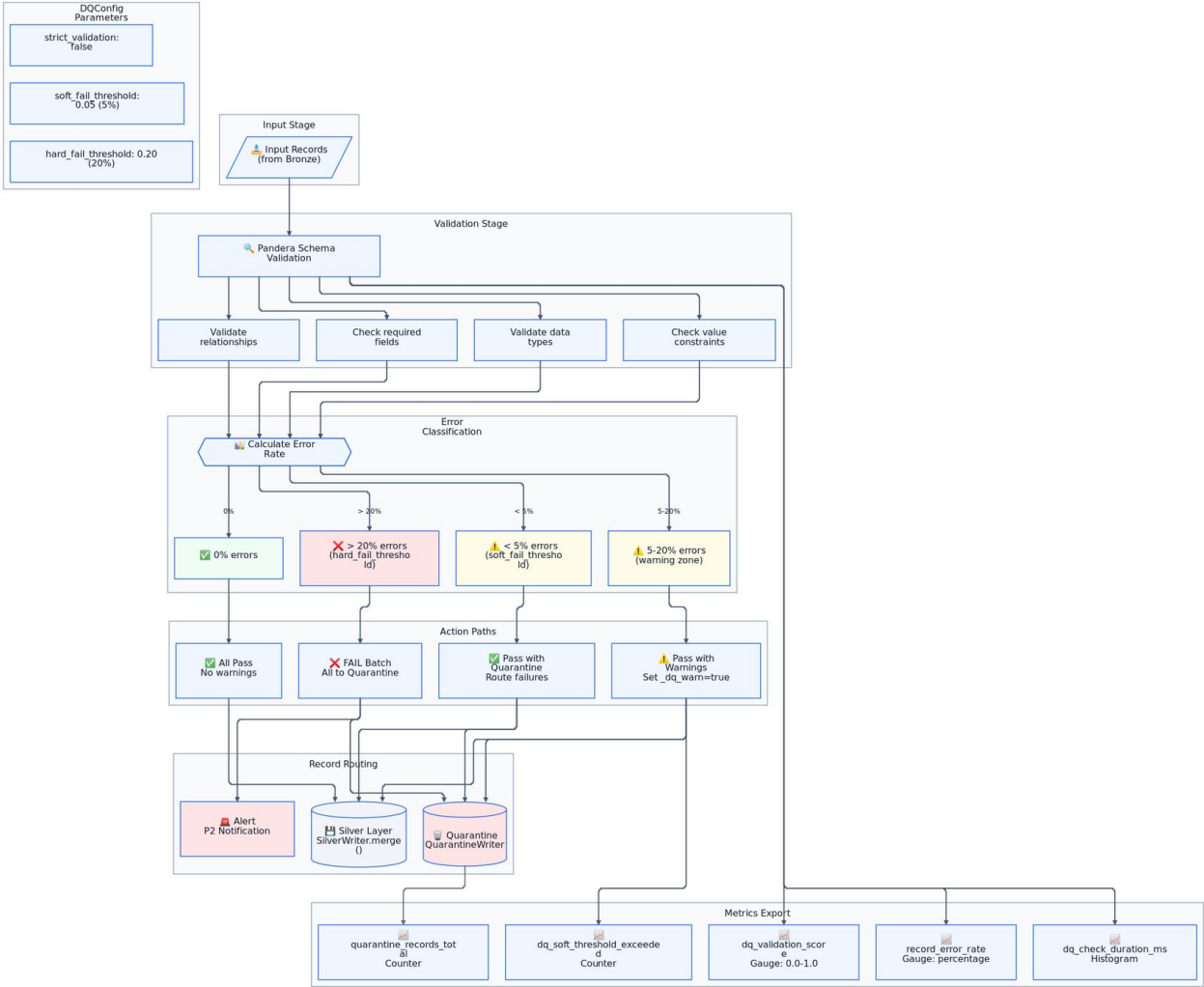
ID: 14-provider-health-states Type: stateDiagram Level: Mixed (System / Component / Class) Date: 2026-02-24



21. Title: Data Quality Check Workflow

ID: 15-dq-check-workflow Type: flowchart Level: Mixed (System / Component / Class) Date: 2026-02-24

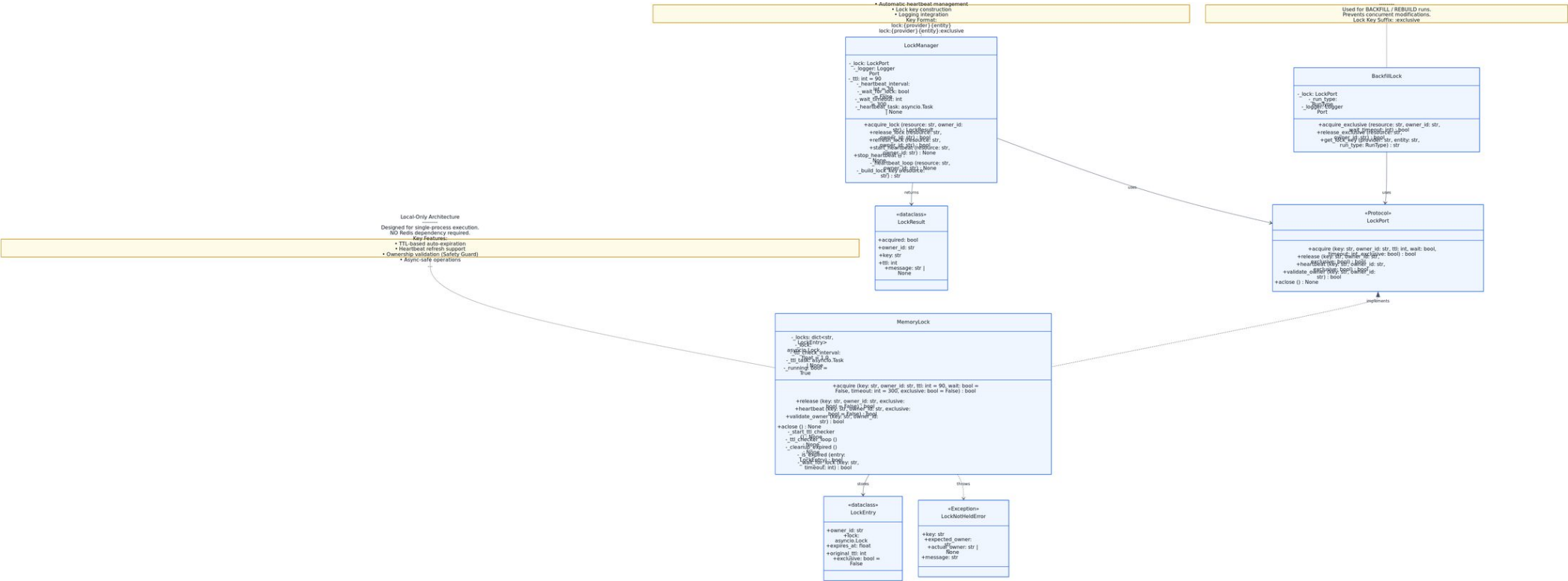
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22. Title: MemoryLock Class Diagram

ID: 16-memory-lock-class Type: classDiagram Level: Mixed (System / Component / Class) Date: 2026-02-24

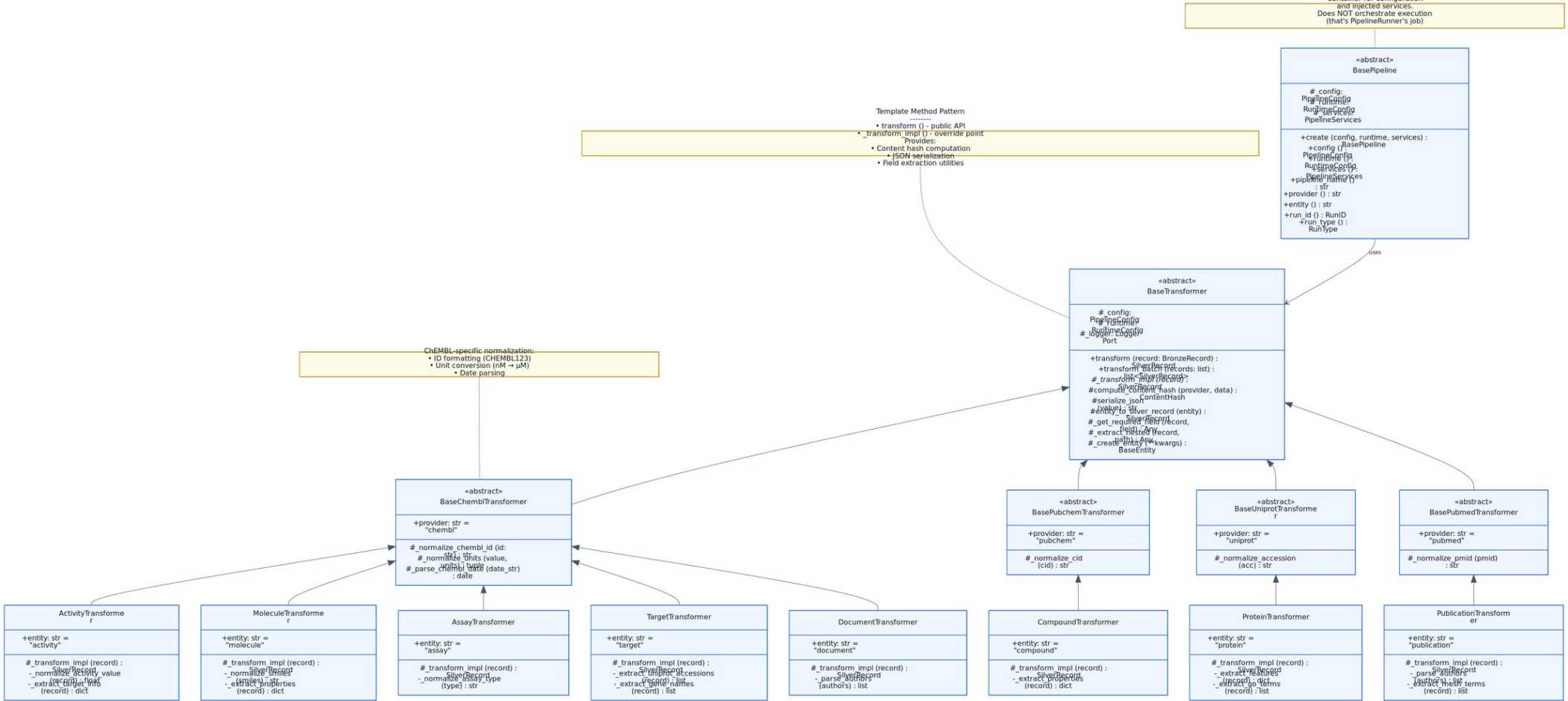
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23. Title: Pipeline and Transformer Class Hierarchy

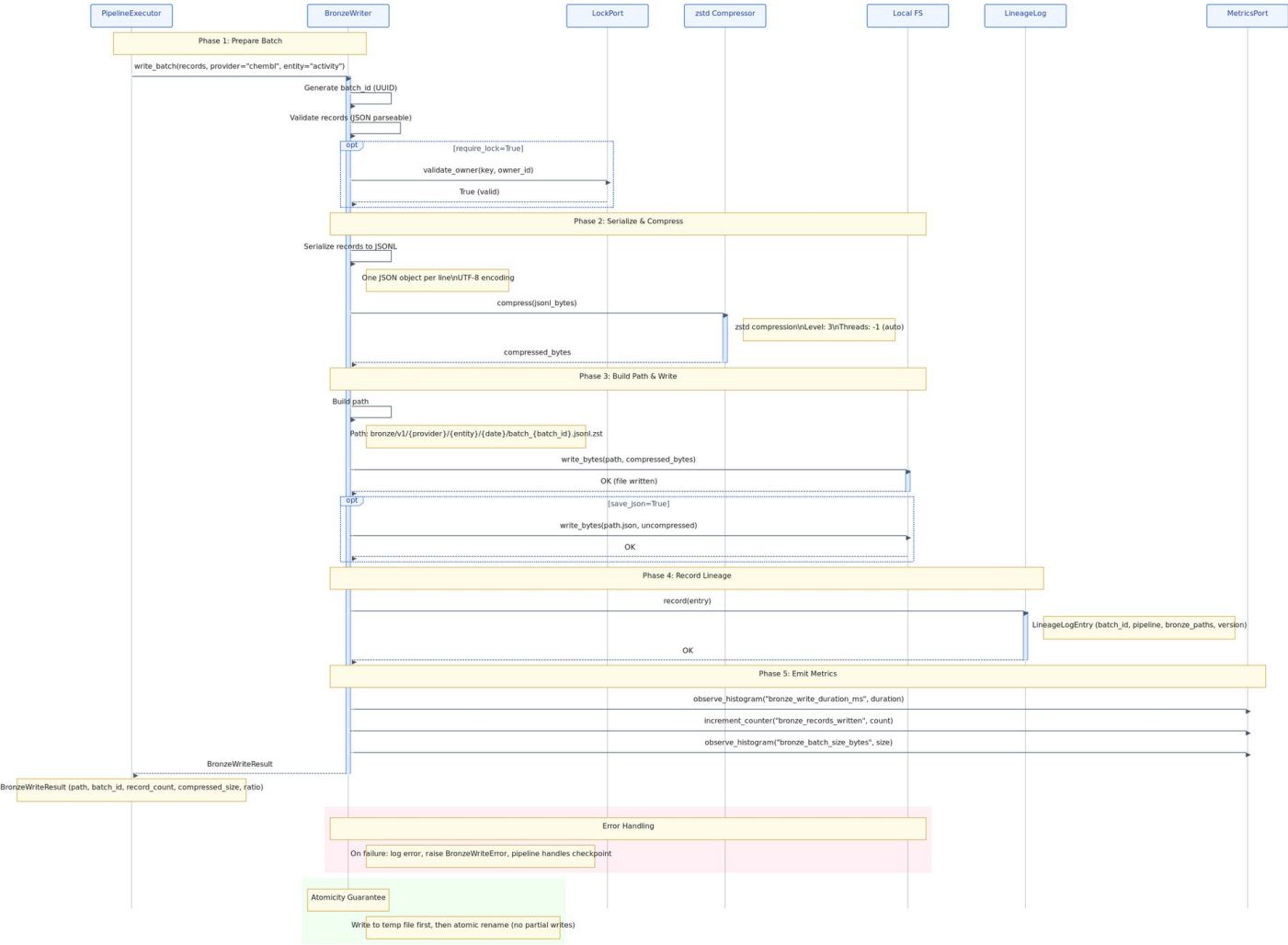
ID: 17-pipeline-hierarchy Type: classDiagram Level: Mixed (System / Component / Class) Date: 2026-02-24

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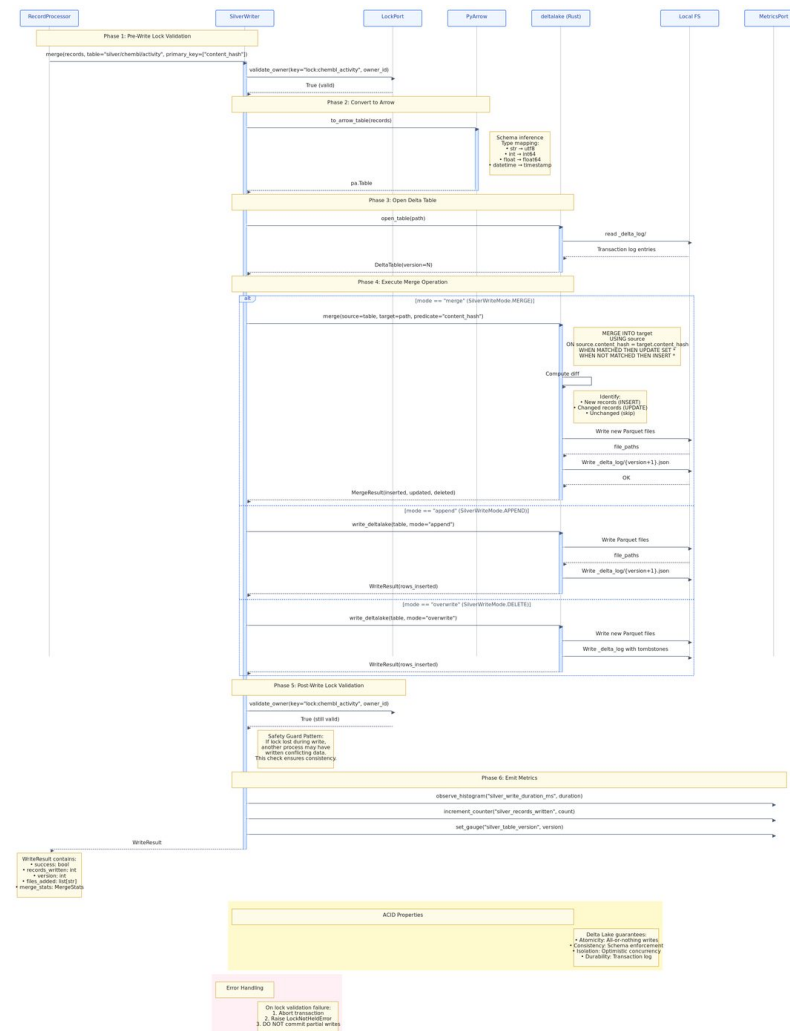
24. Title: Bronze Write Sequence (JSONL + zstd)

ID: 18-bronze-write-sequence Type: sequenceDiagram Level: Mixed (System / Component / Class) Date: 2026-02-24



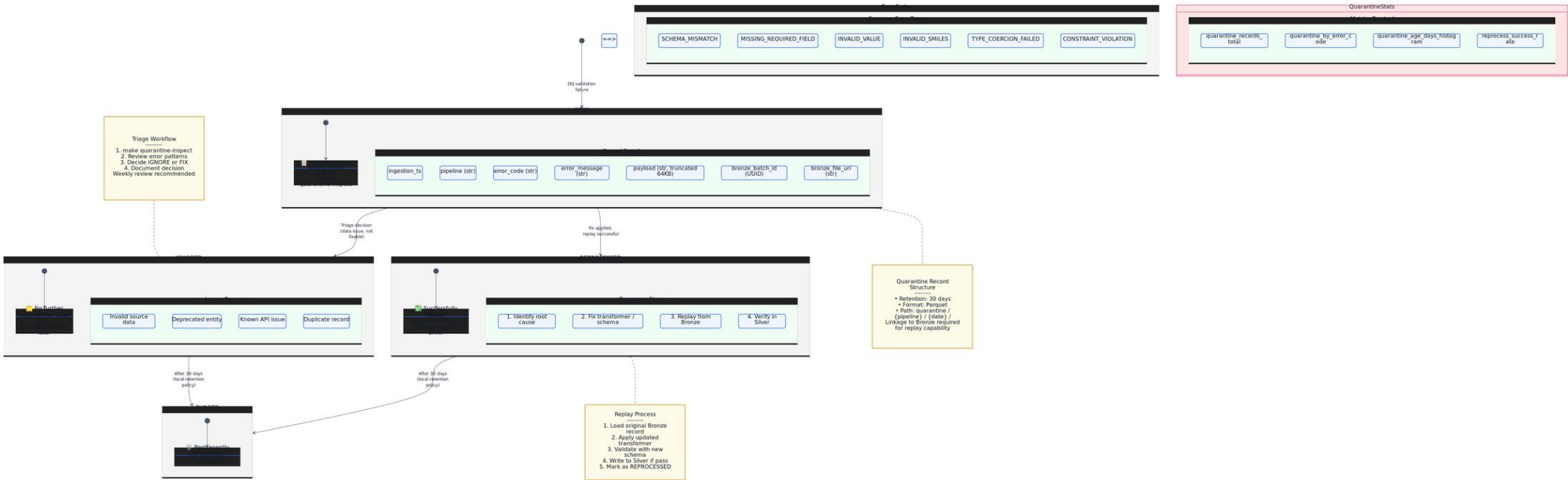
25. Title: Delta Lake Write Sequence (Silver Layer)

ID: 19-delta-lake-write-sequence Type: sequenceDiagram Level: Mixed (System / Component / Class) Date: 2026-02-24



26. Title: Quarantine Record State Machine

ID: 20-quarantine-record-states Type: stateDiagram Level: Mixed (System / Component / Class) Date: 2026-02-24



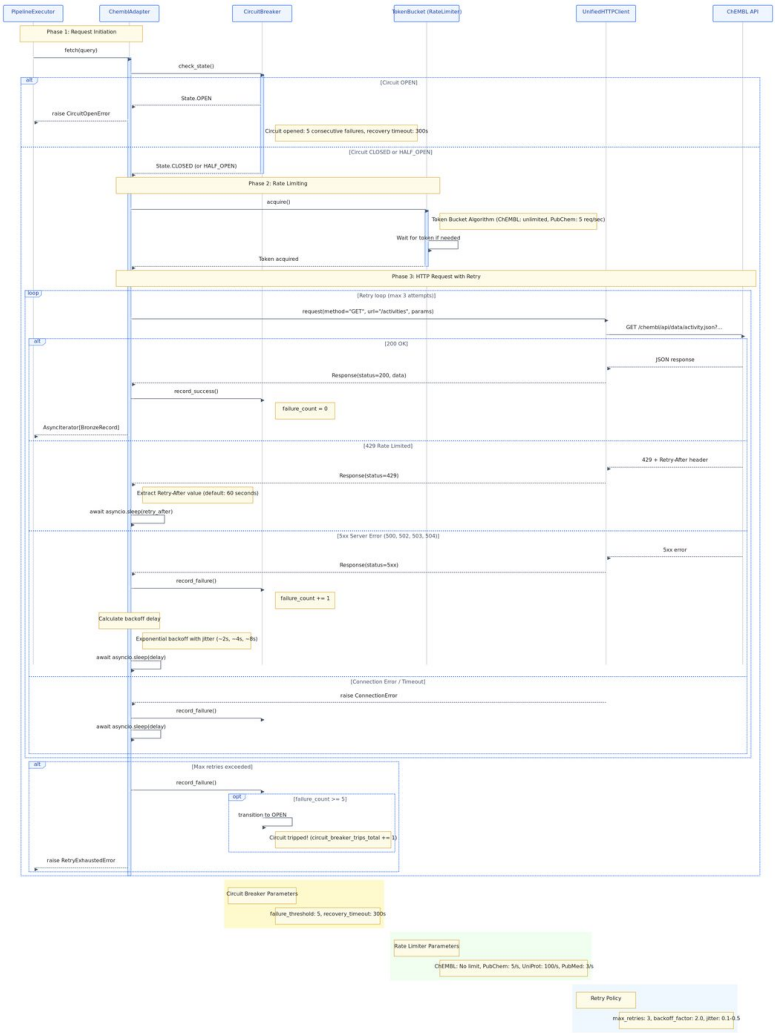
27. Title: Activity Entity Data Flow (Extract → Transform → Load)

ID: 21-activity-entity-data-flow Type: flowchart Level: Mixed (System / Component / Class) Date: 2026-02-24



28. Title: HTTP Client API Request Sequence

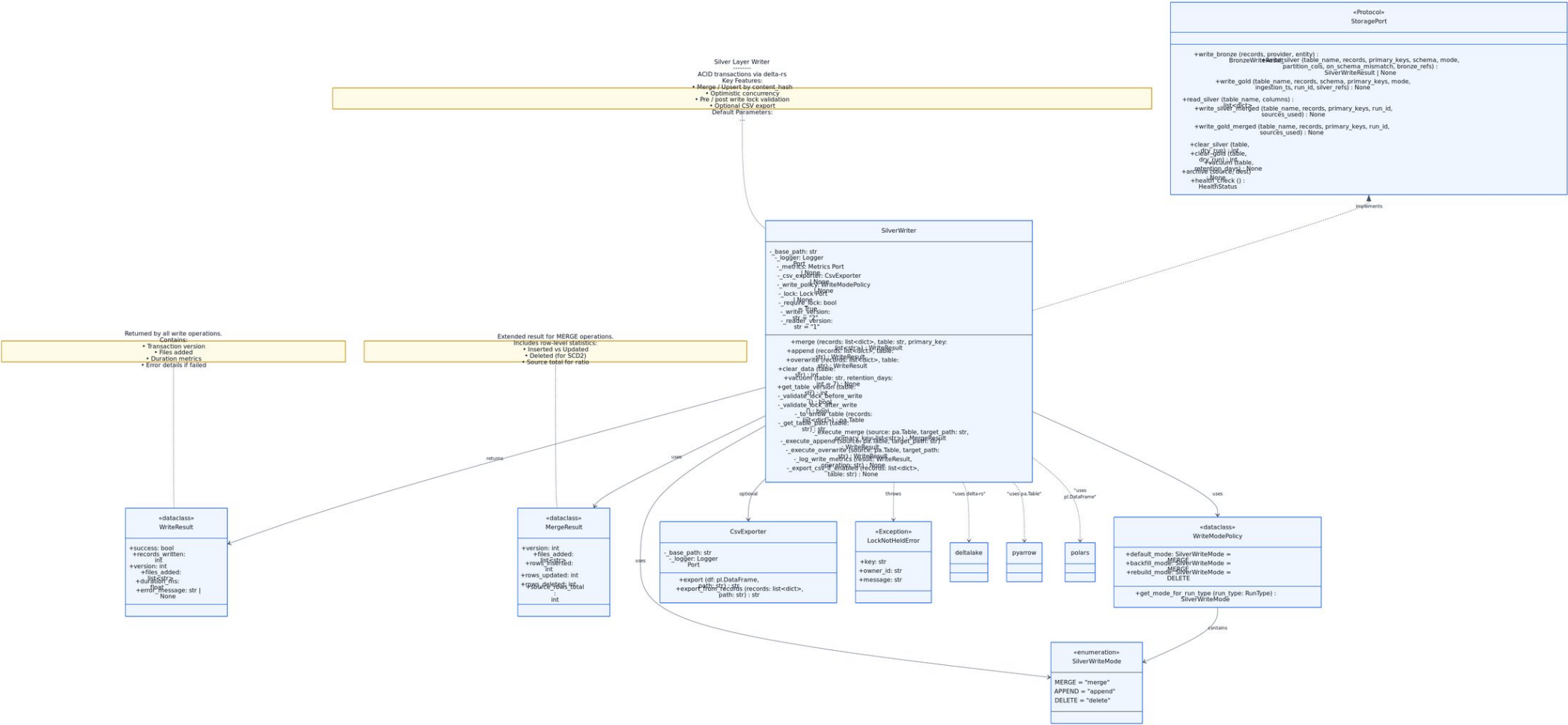
ID: 22-client-api-request-sequence Type: sequenceDiagram Level: Mixed (System / Component / Class) Date: 2026-02-24



29. Title: SilverWriter Class Diagram

ID: 23-silver-writer-class Type: classDiagram Level: Mixed (System / Component / Class) Date: 2026-02-24

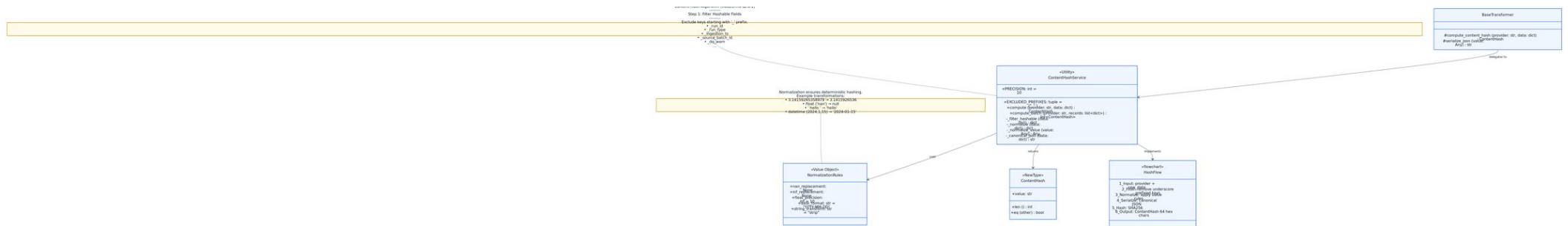
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30. Title: ContentHashService Class Diagram

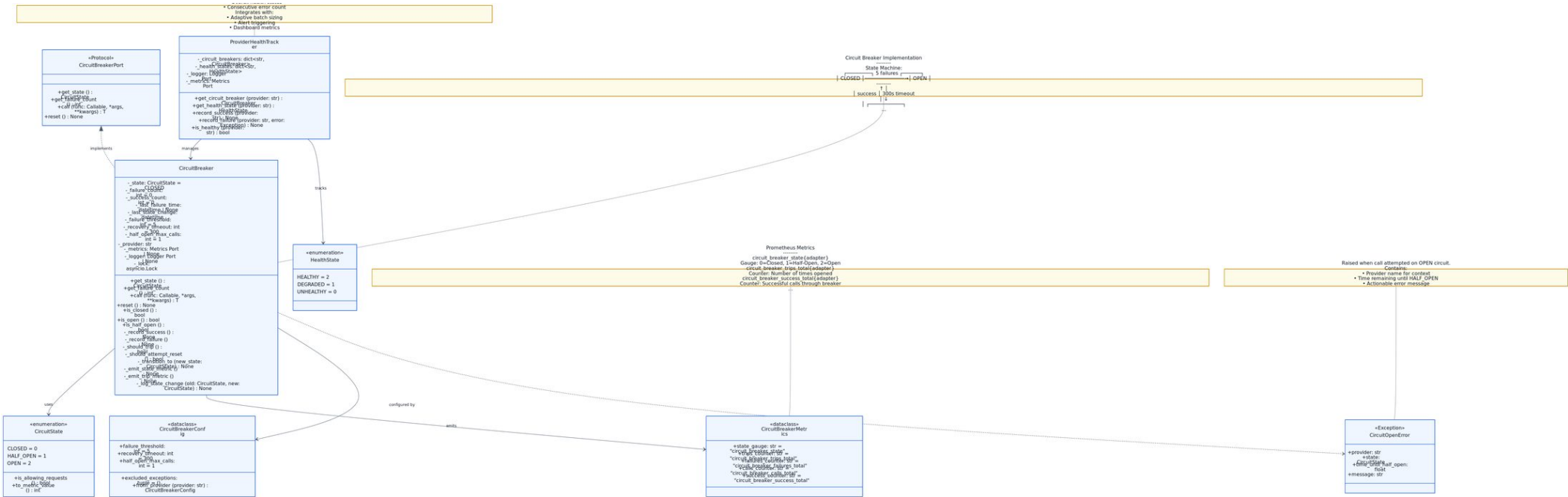
ID: 24-hash-service-class Type: classDiagram Level: Mixed (System / Component / Class) Date: 2026-02-24

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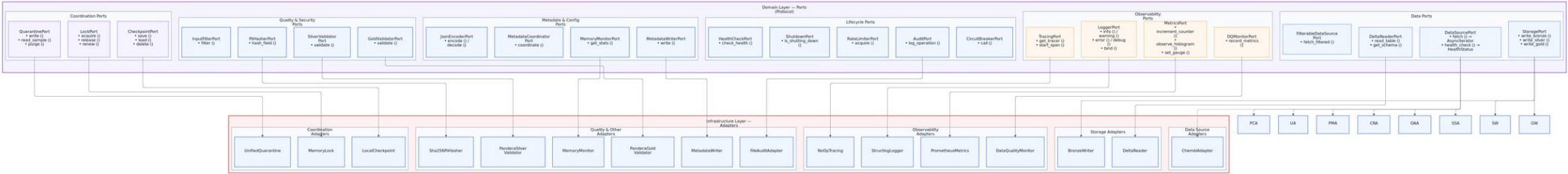
31. Title: Circuit Breaker and Observer Classes

ID: 25-circuit-breaker-observer-class Type: classDiagram Level: Mixed (System / Component / Class) Date: 2026-02-24



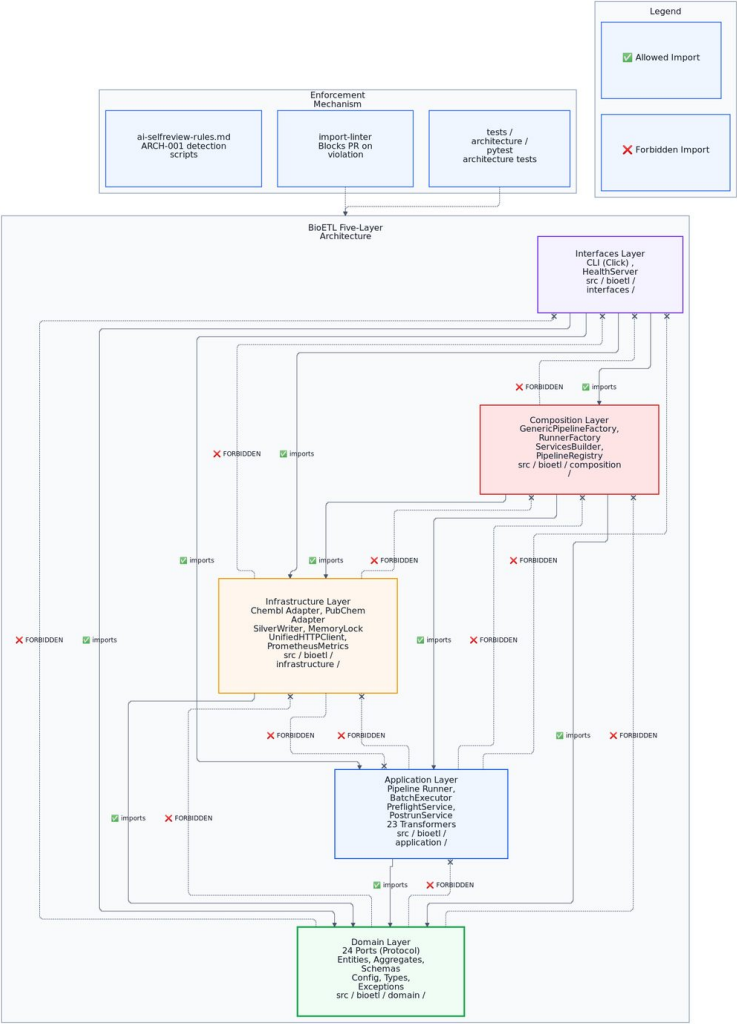
32. Title: Hexagonal Architecture — Ports and Adapters Overview

ID: 26-hexagonal-ports-adapters Type: flowchart Level: Mixed (System / Component / Class) Date: 2026-02-24



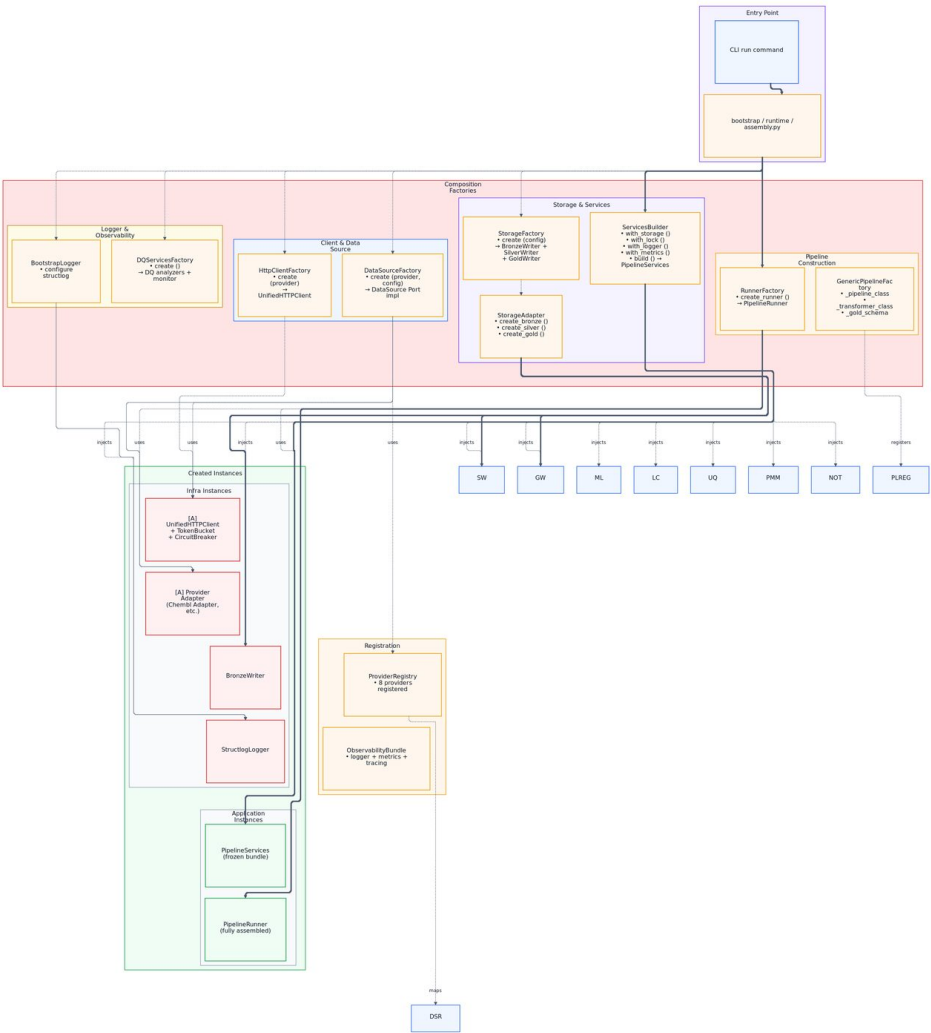
33. Title: Five-Layer Import Matrix Enforcement (ARCH-001)

ID: 27-import-matrix-enforcement Type: flowchart Level: Mixed (System / Component / Class) Date: 2026-02-24



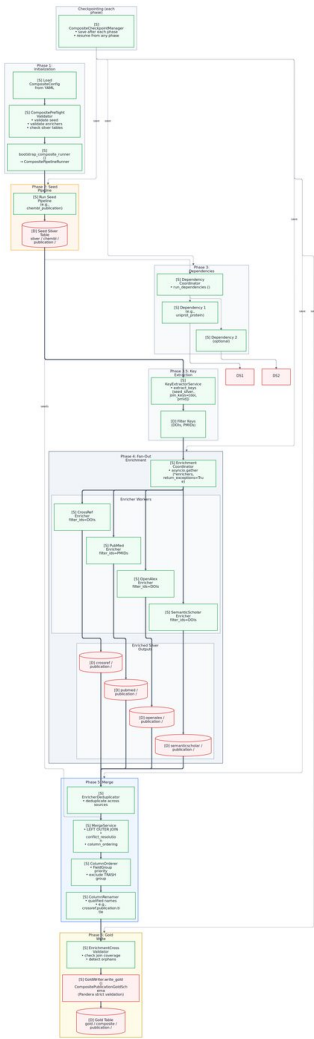
34. Title: Composition Root Wiring — Full DI Graph

ID: 28-composition-root-di-graph Type: flowchart Level: Mixed (System / Component / Class) Date: 2026-02-24



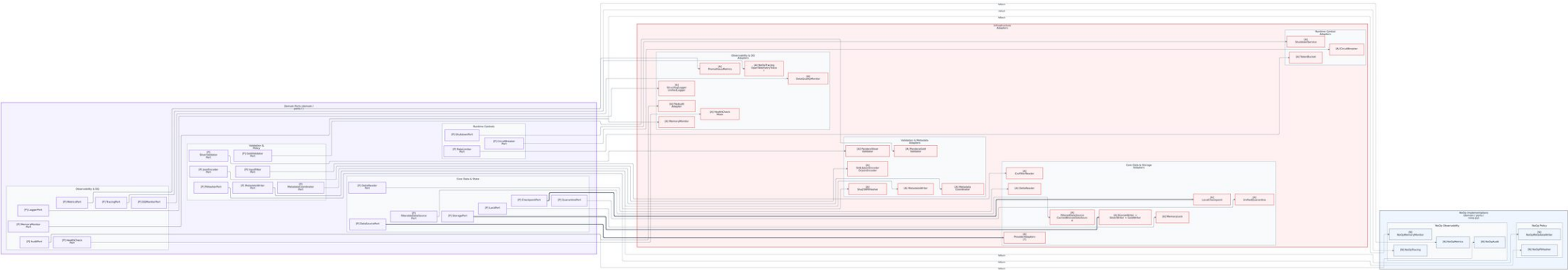
35. Title: Composite Pipeline Full Workflow — Seed to Gold (ADR-026)

ID: 29-composite-pipeline-workflow Type: flowchart Level: Mixed (System / Component / Class) Date: 2026-02-24



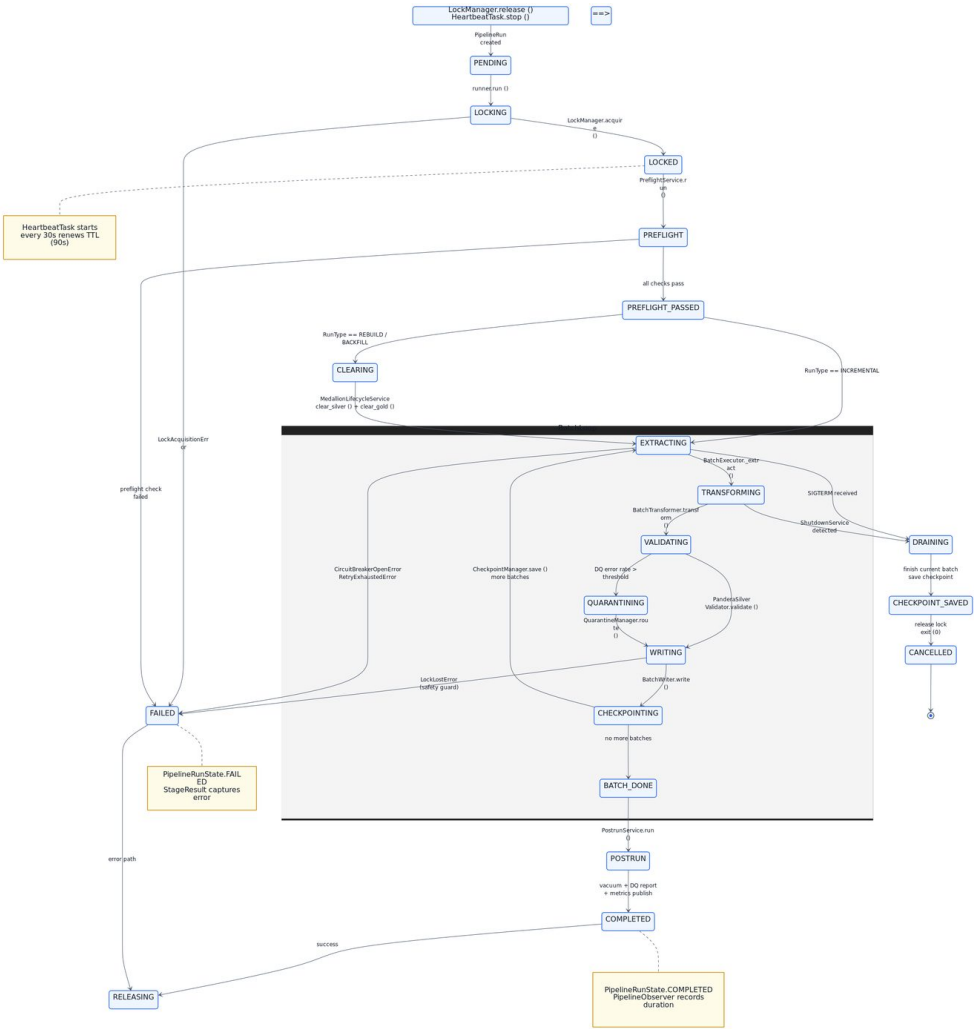
36. Title: Port-to-Adapter Mapping Table Diagram

ID: 30-port-adapter-mapping Type: flowchart Level: Mixed (System / Component / Class) Date: 2026-02-27



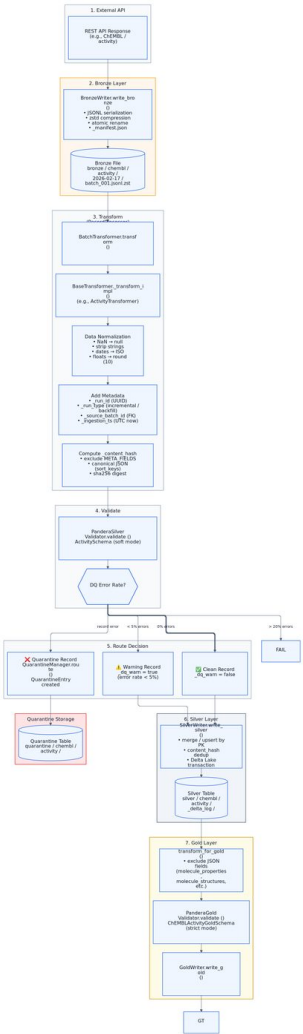
37. Title: Pipeline Run Lifecycle — From Config to Completion

ID: 31-pipeline-run-lifecycle Type: stateDiagram Level: Mixed (System / Component / Class) Date: 2026-02-24



38. Title: Record Processing Pipeline — Single Record Journey

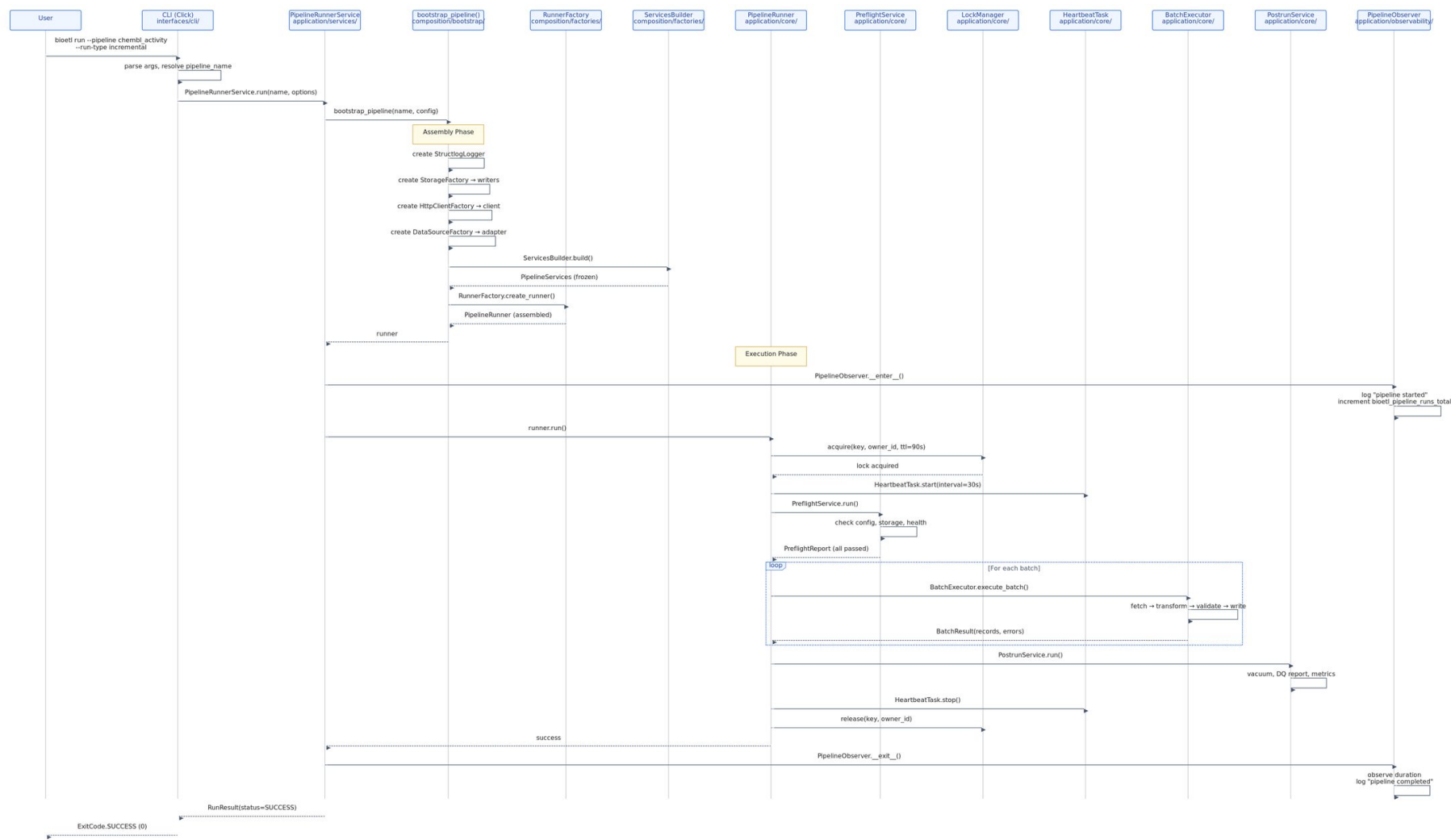
ID: 32-single-record-journey Type: flowchart Level: Mixed (System / Component / Class) Date: 2026-02-24



39. Title: CLI Run Command → PipelineRunner Full Interaction

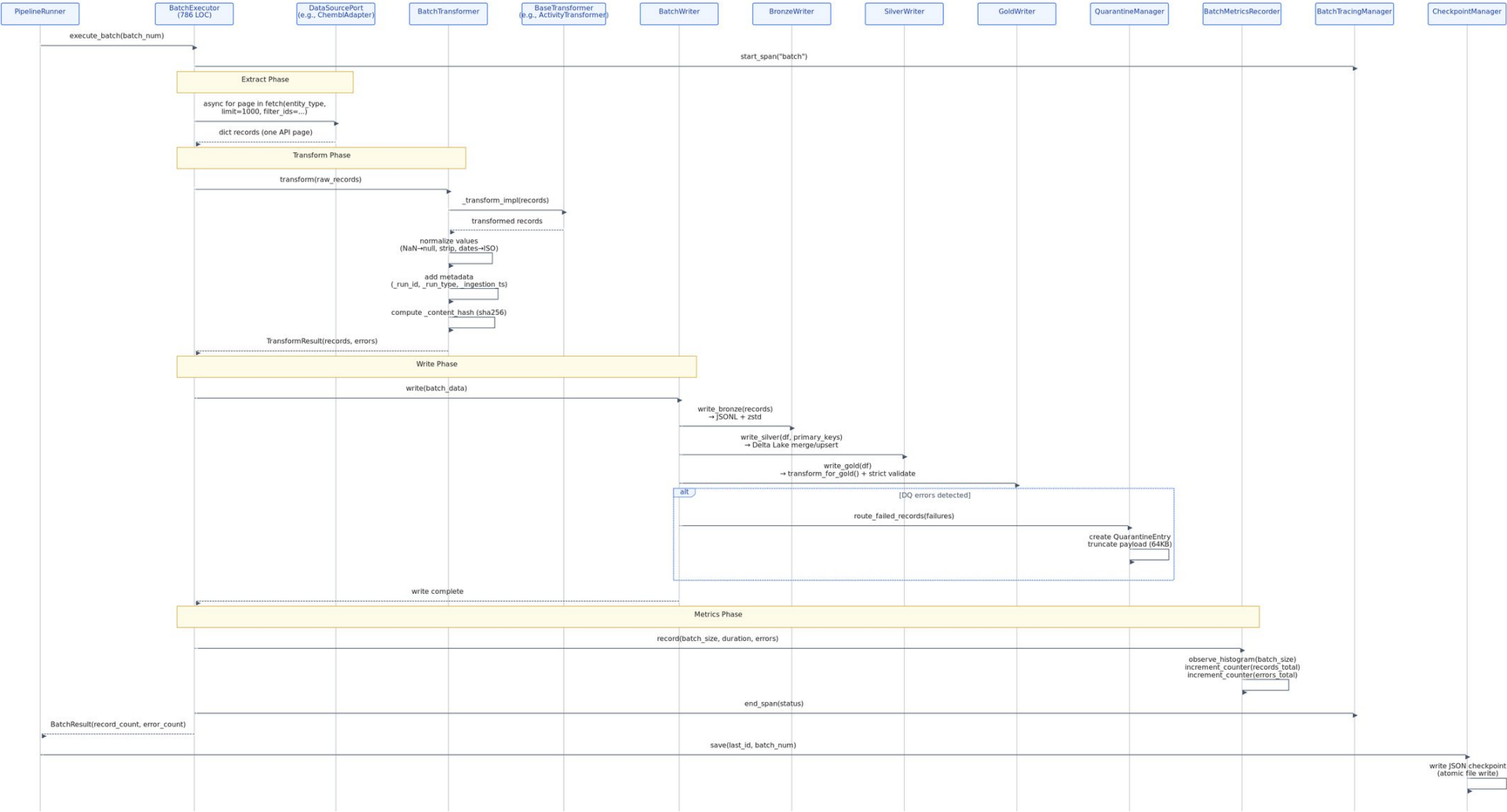
ID: 33-cli-run-interaction Type: sequenceDiagram Level: Mixed (System / Component / Class) Date: 2026-02-24

Source: docs/02-architecture/mmd-diagrams/foundation/33-cli-run-interaction.mmd | Description: docs/02-architecture/diagram-descriptions/mmd-diagrams/foundation/33-cli-run-interaction.md



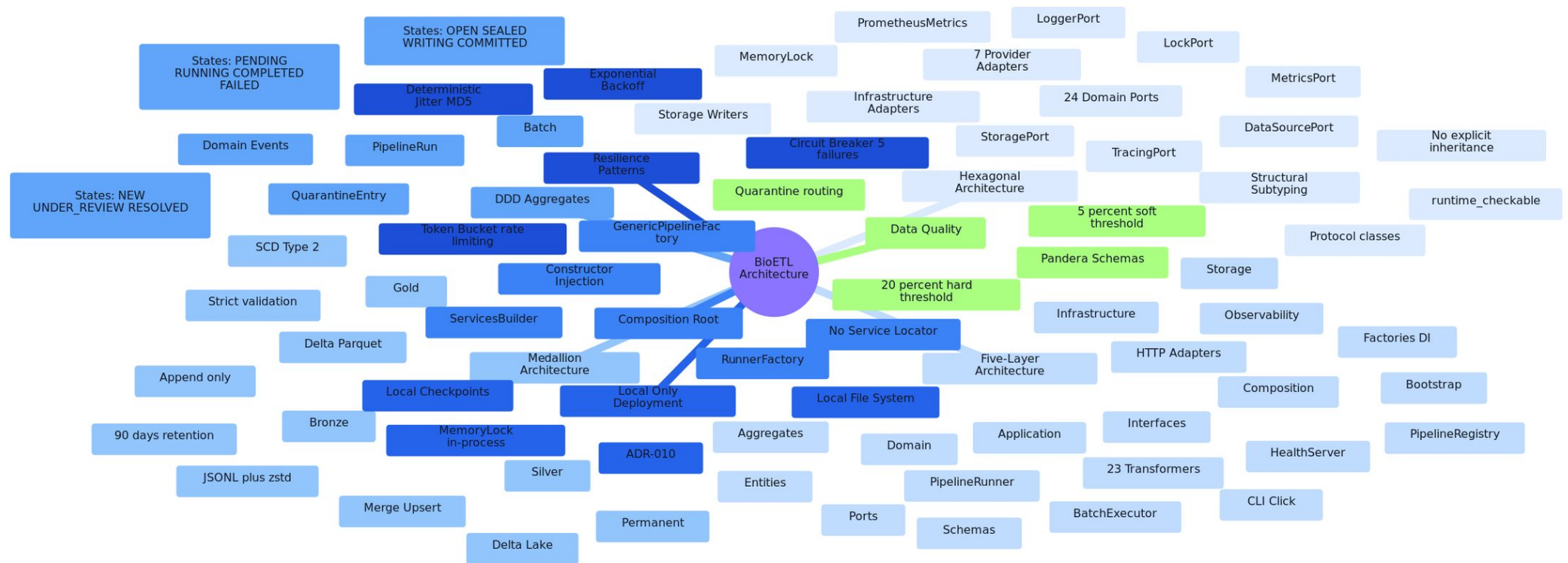
40. Title: Batch Processing Flow — Extract to Write

ID: 34-batch-processing-flow Type: sequenceDiagram Level: Mixed (System / Component / Class) Date: 2026-02-24



41. Title: Architecture Principles Mind Map

ID: 36-architecture-principles-mindmap Type: mindmap Level: Mixed (System / Component / Class) Date: 2026-02-24



42. Title: CLI Entry Point to Pipeline Execution Full Chain

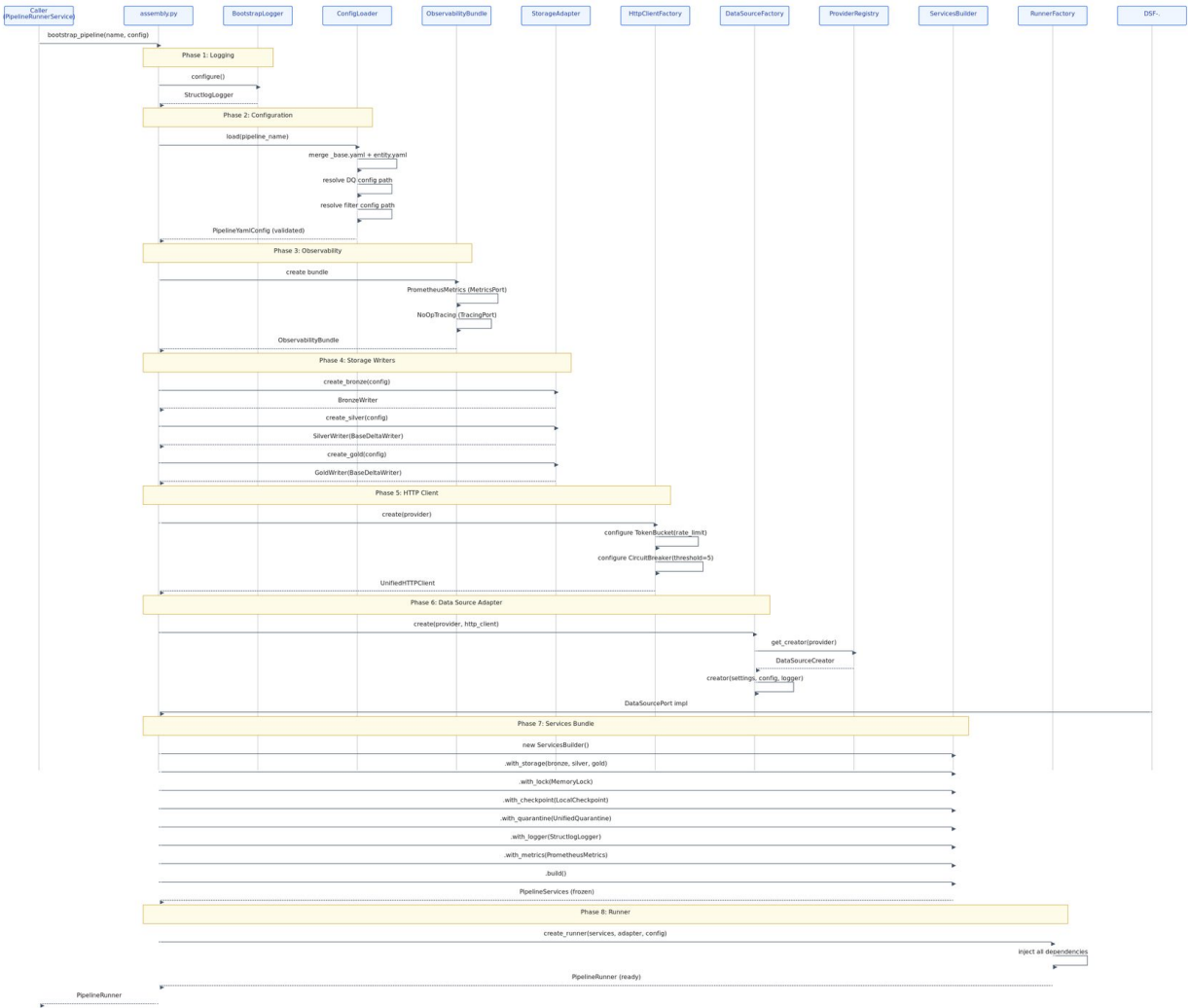
ID: 37-cli-entry-full-chain Type: flowchart Level: Mixed (System / Component / Class) Date: 2026-02-24

Source: docs/02-architecture/mmd-diagrams/foundation/37-cli-entry-full-chain.mmd | Description: docs/02-architecture/diagram-descriptions/mmd-diagrams/foundation/37-cli-entry-full-chain.md



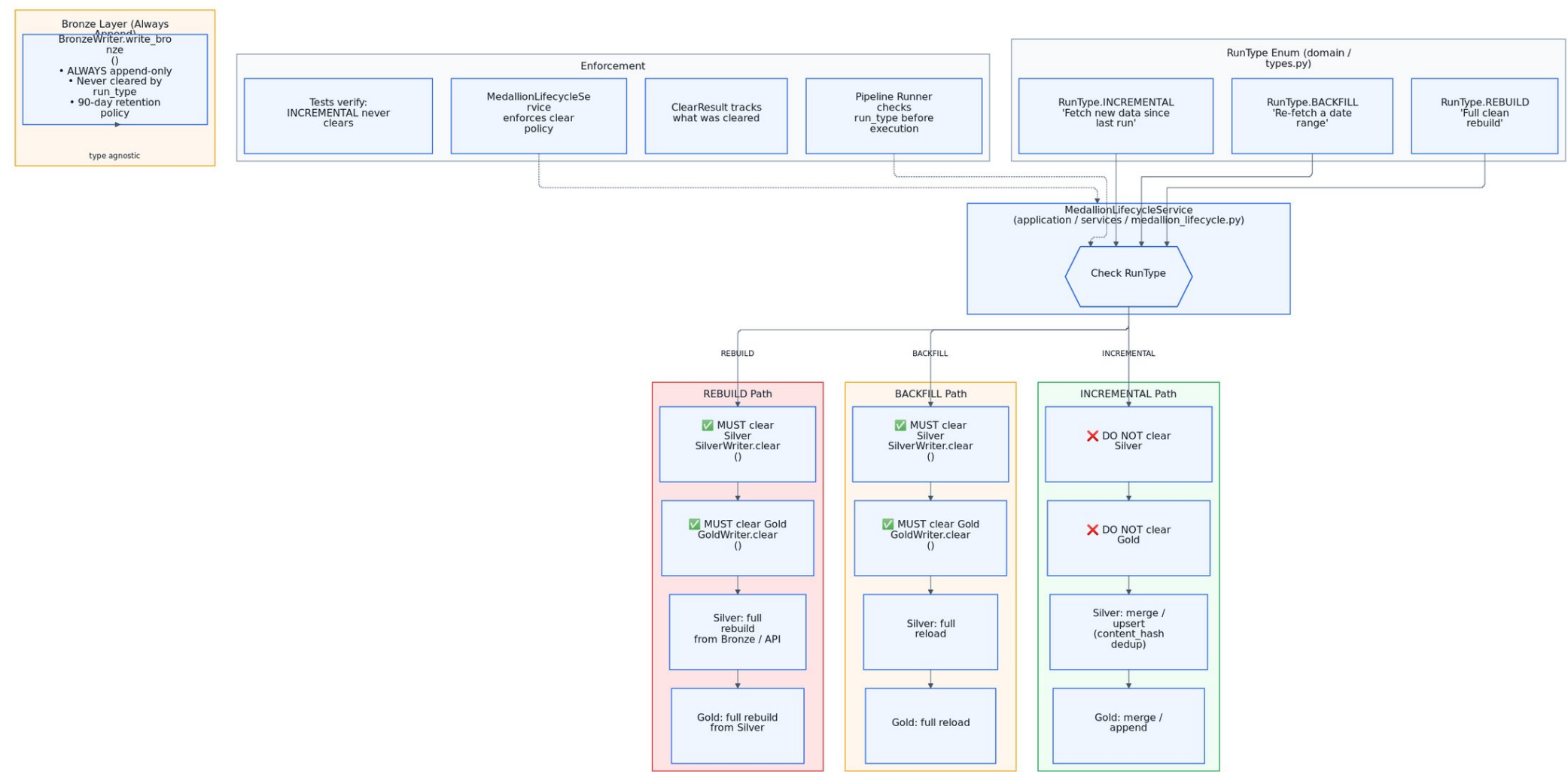
43. Title: Runtime Assembly Sequence — bootstrap/runtime/assembly.py

ID: 38-runtime-assembly-sequence Type: sequenceDiagram Level: Mixed (System / Component / Class) Date: 2026-02-24



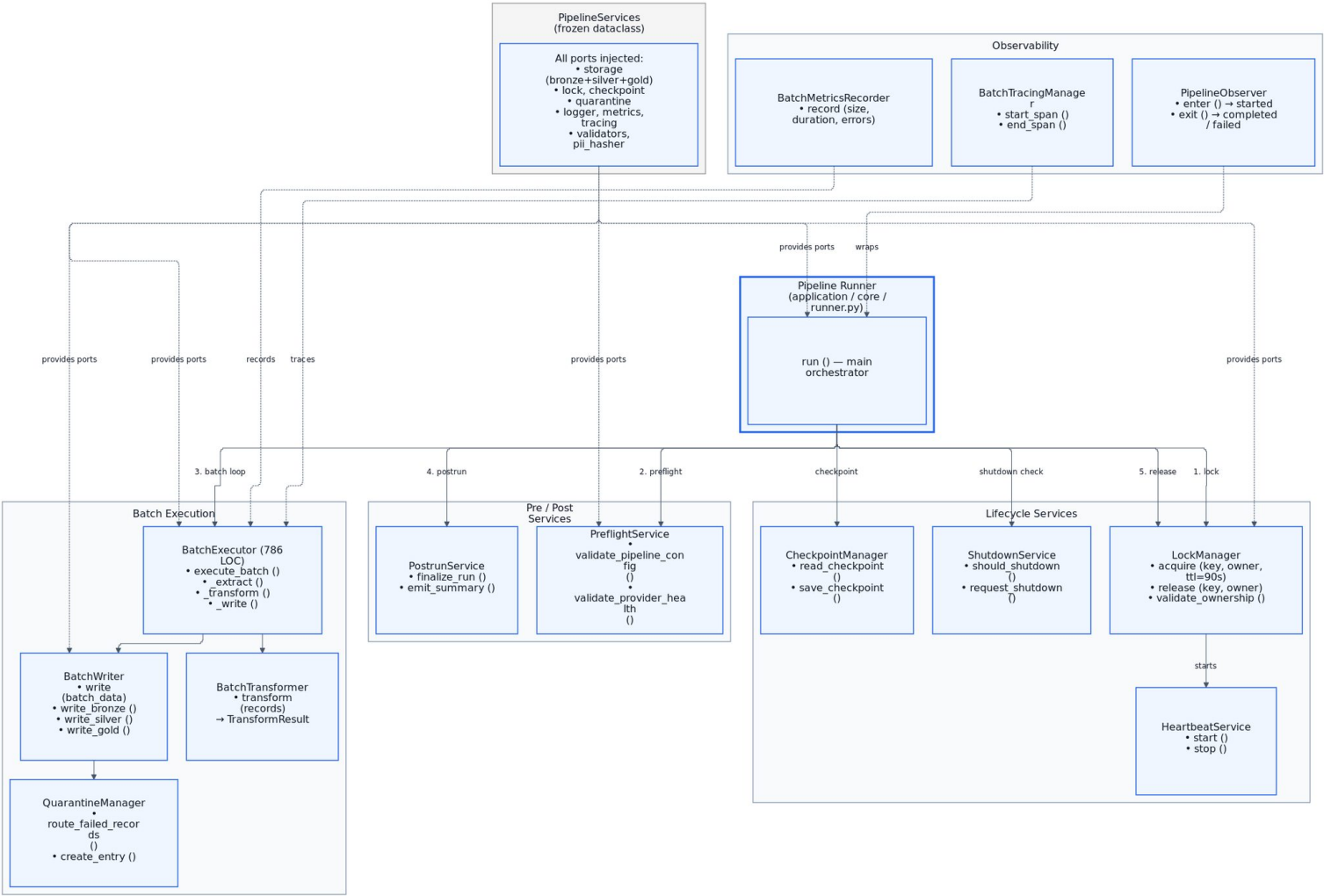
44. Title: Medallion Architecture Invariants (ARCH-007)

ID: 39-medallion-invariants Type: flowchart Level: Mixed (System / Component / Class) Date: 2026-02-24



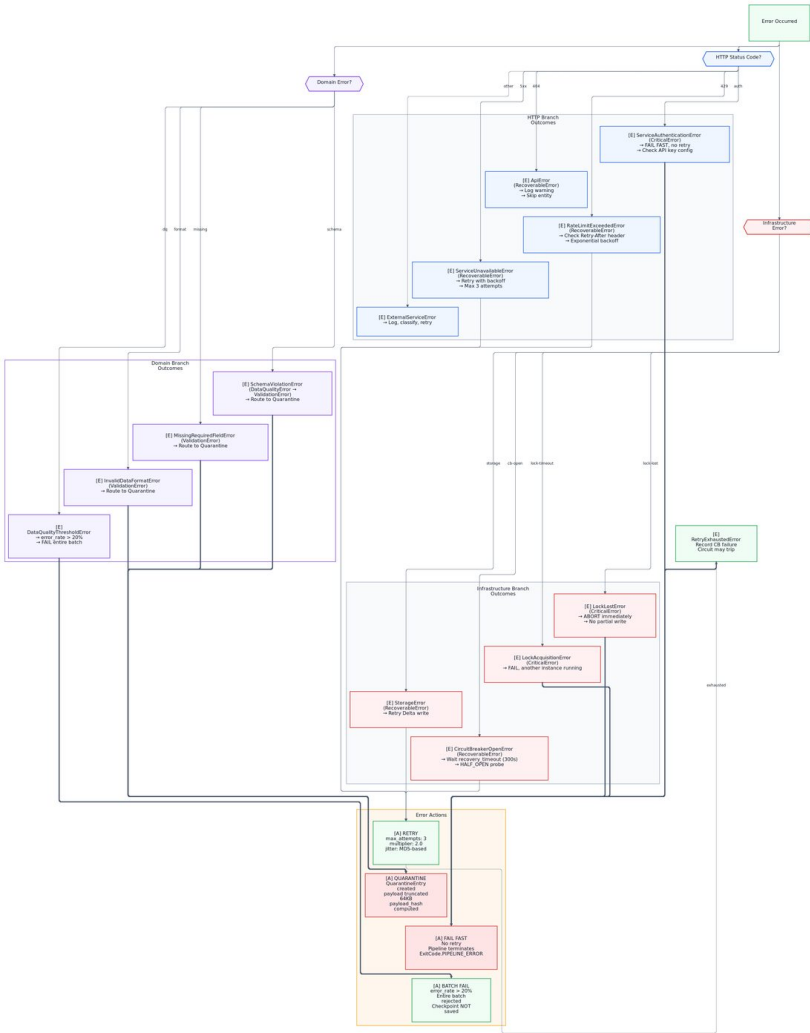
45. Title: Application Core Component Collaboration

ID: 40-application-core-collaboration Type: flowchart Level: Mixed (System / Component / Class) Date: 2026-02-24



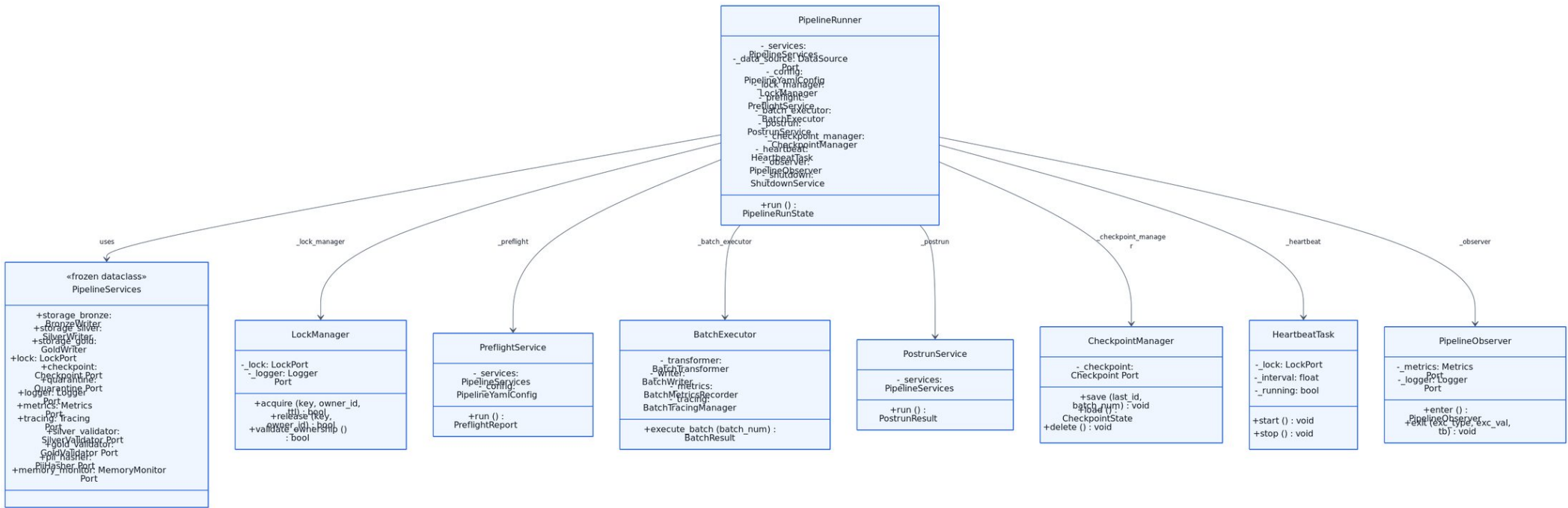
46. Title: Error Classification Decision Tree — Full Logic

ID: 41-error-classification-tree Type: flowchart Level: Mixed (System / Component / Class) Date: 2026-02-24



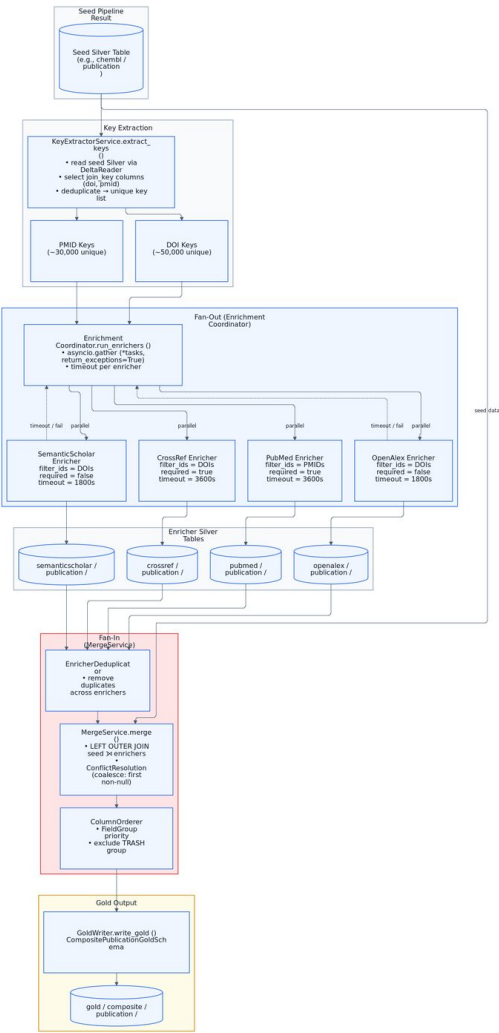
47. Title: PipelineRunner Internal Component Diagram

ID: 42-pipeline-runner-class Type: classDiagram Level: Mixed (System / Component / Class) Date: 2026-02-24



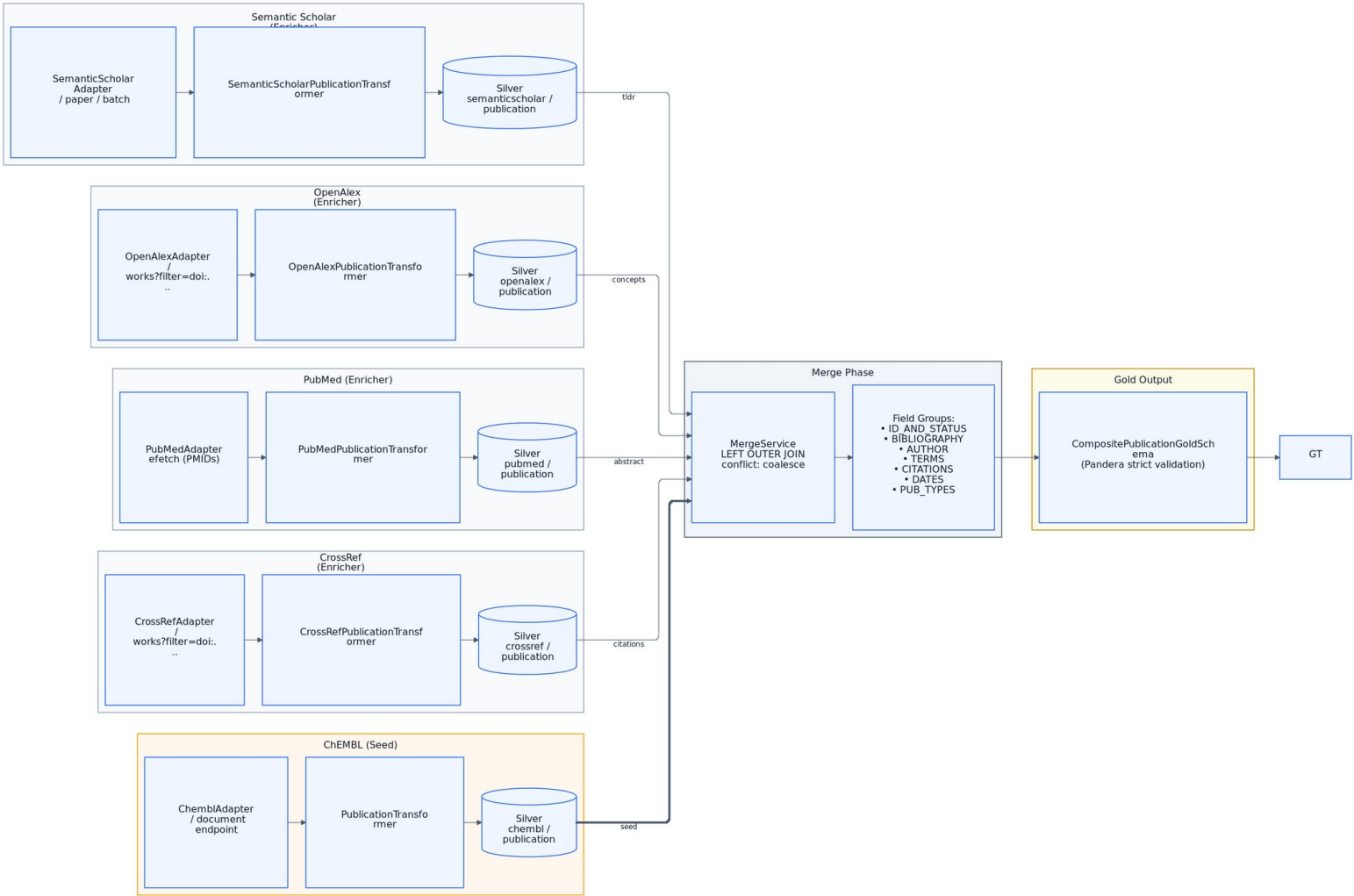
48. Title: Fan-Out/Fan-In Pattern — Composite Pipeline Enrichment

ID: 43-fan-out-fan-in-pattern Type: flowchart Level: Mixed (System / Component / Class) Date: 2026-02-24



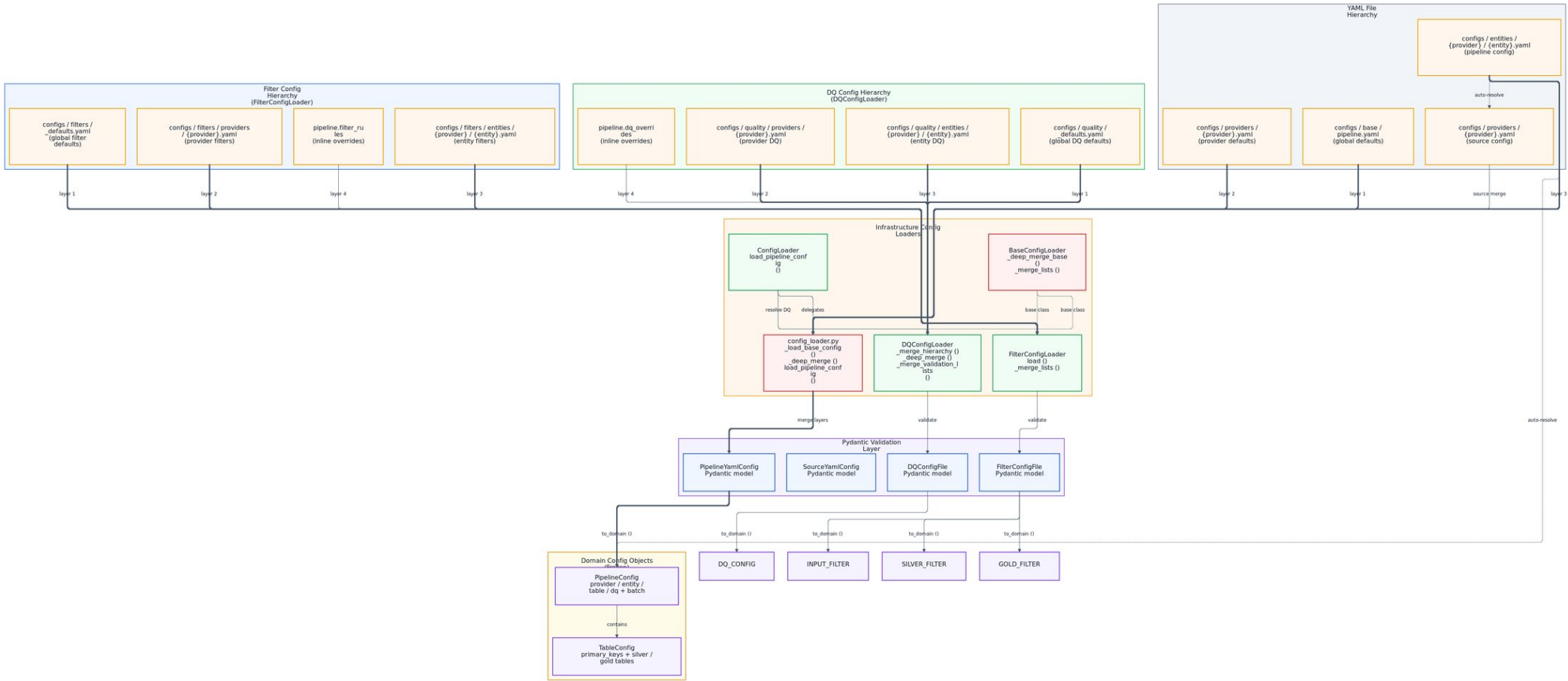
49. Title: Cross-Provider Data Enrichment Flow — Publication

ID: 44-cross-provider-enrichment Type: flowchart Level: Mixed (System / Component / Class) Date: 2026-02-24



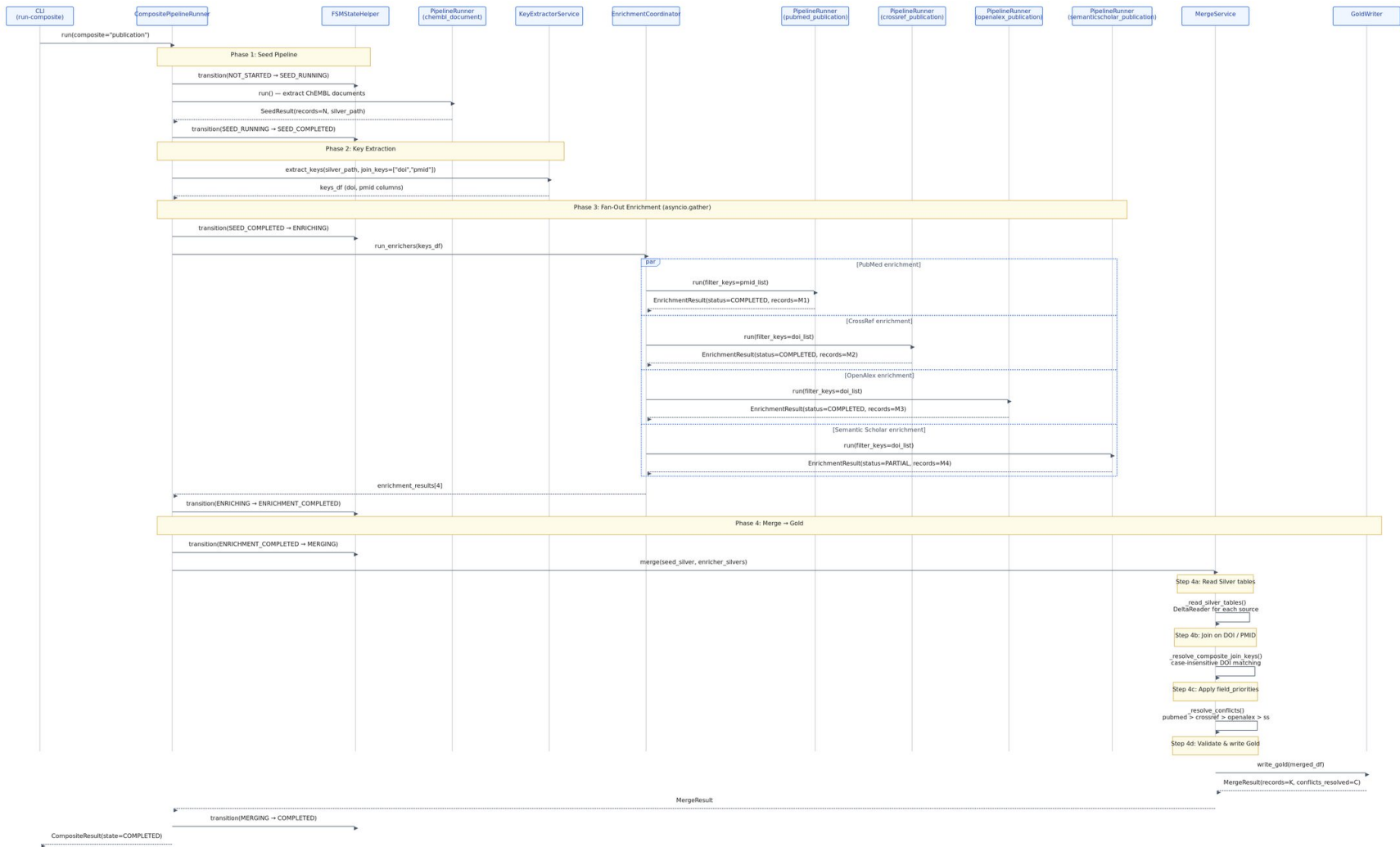
50. Title: YAML Configuration Resolution Chain

ID: 46-yaml-config-resolution Type: flowchart Level: Mixed (System / Component / Class) Date: 2026-02-27



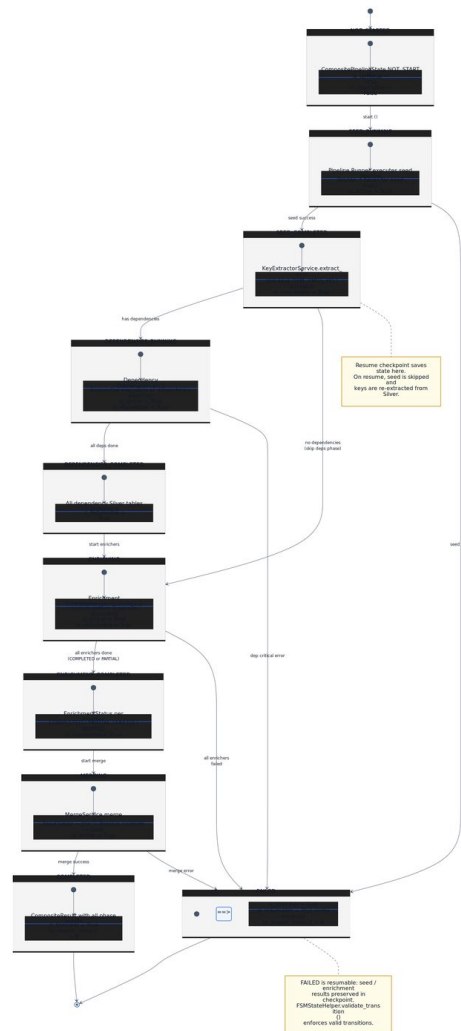
51. Title: Publication Composite — Merge All Sources

ID: 47-publication-merge-sources Type: sequenceDiagram Level: Mixed (System / Component / Class) Date: 2026-02-24



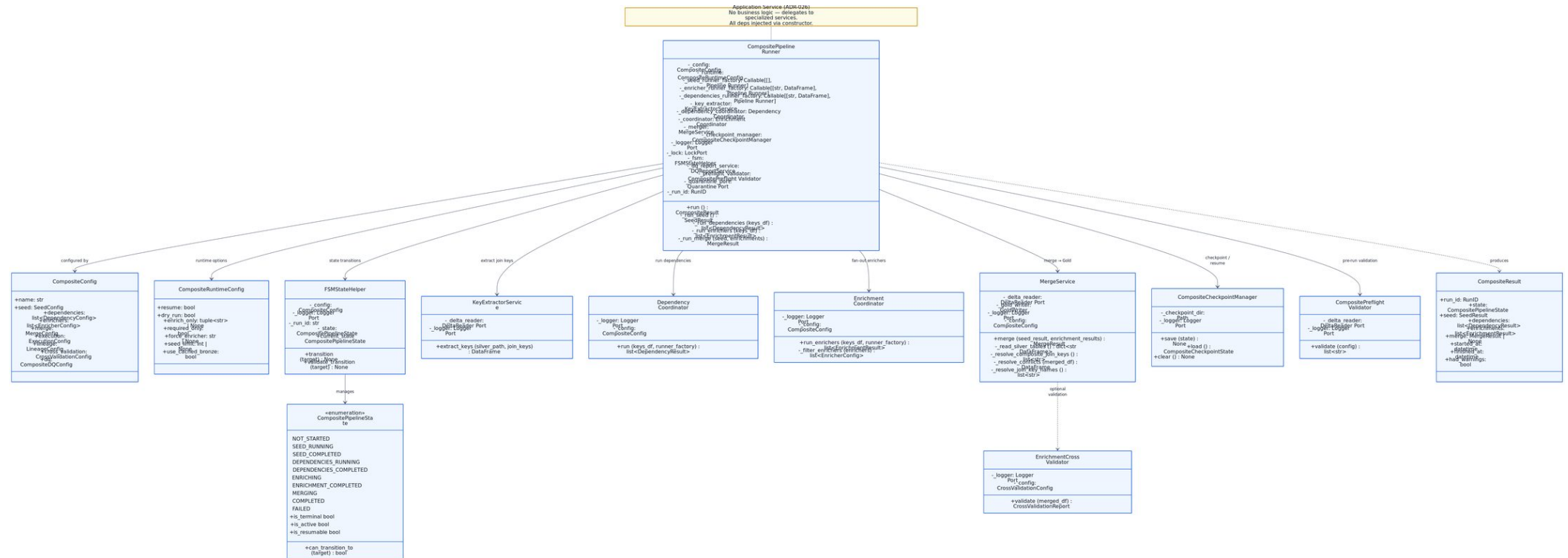
52. Title: Composite Pipeline Phase Lifecycle (FSM)

ID: 48-composite-phase-lifecycle Type: stateDiagram Level: Mixed (System / Component / Class) Date: 2026-02-24



53. Title: CompositePipelineRunner — Component Diagram

ID: 49-composite-runner-class Type: classDiagram Level: Mixed (System / Component / Class) Date: 2026-02-24



54. Title: Exception Hierarchy — Full Tree

ID: 50-exception-hierarchy Type: flowchart Level: Mixed (System / Component / Class) Date: 2026-02-24

